Gradient descent on Boolean targets as noise-limited greedy feature growth: experiments, discrete models, and generalization bounds

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Abstract

We look for a low-level, discrete description of what gradient descent does when it trains a small network in the feature-learning regime, analogous to the correspondence between Bayesian neural networks and Occam circuit learning. The testbed is a μP two-hidden-layer ReLU network trained by Adam on Boolean targets, with input dimension $d \in \{15, 30, 60, 120\}$. We report three things.

Phenomenology. The training-set size n^* at which test error reaches 0.1 grows polynomially in d, with an exponent set by how many inputs must be found at once. For k-bit parities it grows like $d^{1.0}$, $d^{1.9}$ and $d^{2.4}$ for $k = 2, 3, 4$. This is close to the correlational-SQ rate d^{k-1} for $k \leq 3$, and far from the $k \log d$ an Occam learner needs. The ratio of the networks' n^* to an Occam learner's grows with the target's leap, from about $8\times$ at leap 1 to $85\times$ at leap 4.

A discrete model. A greedy feature-growth model, in which a candidate feature is accepted once its correlation with the residual beats the sampling noise of a continuous search for it, reproduces the networks' n^* to within about $1.15\times$ on 20 random formulas it was not calibrated on. Ablations show that this noise, not the model's race dynamics, sets n^*. A two-constant closed form for n^* predicted the networks' 3-parity sample complexity at $d = 120$ to within $1.2\times$ before those networks were trained; the simulation model was $3\times$ too optimistic there. On partially learned tasks, the networks' individual mistakes look like those of their initial kernel. About 45% of the output variance is kernel-shaped fitted sample noise, which suggests the picture "random-feature kernel background plus a few grown features".

Bounds. We prove versions of "short training ⇒ small circuit ⇒ generalization" for:

- the discrete model, with size read as depth, incoherence-weighted size, or readout norm;
- diagonal linear networks;
- two-layer quadratic networks with fixed output signs, where the number of grown units is at most $\kappa(\tau/\Theta)^2$ uniformly in time;
- two-layer ReLU networks under gradient flow, through the first escape from the initial saddle.

On the real networks, a-priori bounds are vacuous or weak. Distillation certificates ("the network agrees on unlabelled inputs with a small circuit") give non-vacuous bounds on 221 of 315 trained networks, and are never violated.

*Written by an AI system in a series of sessions prompted and steered by Kaarel Hänni. Not reviewed by a human. The proofs were checked by independent adversarial AI referees, and the numbers by independent AI fact-checkers, as described in Appendix D. Code, data, derivations and referee reports are on the `claude/nifty-mccarthy-j0lw2a` branch of https://github.com/kaarelh/AI-works, folder `gd-circuit-growth/`.

Finally, we test a question about compositional structure that leap complexity cannot see: does $s + x_1x_2 + x_3x_4 + x_1x_2x_3x_4$ (with $s = x_1 + \cdots + x_4$) learn its degree-4 term faster than $s + x_1x_2x_3x_4$? We recorded predictions before running this test.

Contents

1 Introduction **4**

2 Setup **5**

3 What the networks do **6**
3.1 Sample complexity at $d = 30$ . . . 6
3.2 Dependence on the input dimension . . . 7
3.3 Gradient descent against Occam, by leap . . . 8
3.4 Robustness: optimizer and width . . . 8

4 A discrete model of the feature channel **10**
4.1 The race model . . . 10
4.2 What makes it work: ablations . . . 10
4.3 The noise law and a closed-form predictor . . . 11
4.4 Kernel background plus grown features . . . 13

5 Does compositional structure beyond leap matter? **14**

6 Generalization bounds **14**
6.1 What is proved . . . 14
6.2 What can be certified for the trained networks . . . 15
6.3 What the bounds do not answer . . . 16

7 Discussion **16**
7.1 The picture . . . 16
7.2 Relation to NNBayes ≈ Occam . . . 17
7.3 What is in the literature, and what is new . . . 17
7.4 Leap counts supports, not structure . . . 18
7.5 Limitations . . . 18
7.6 Open problems . . . 18

A Bounds for the discrete circuit-growth model and for diagonal linear networks **20**
A.1 Setting, notation and counting . . . 20
A.2 The Occam bound, uniform over circuit size . . . 21
A.3 The clock race . . . 22
A.4 What training time controls . . . 23
A.5 Test error as a function of training time . . . 25
A.6 SGD and minibatches: reproducing the population run . . . 26
A.7 Greedy growth versus minimum circuits . . . 27
A.8 Diagonal linear networks: a rigorous instance (Section C2) . . . 29
A.9 Conditional transfer to ReLU networks (Section C3) . . . 31

B Bounds for the trained networks **33**
B.1 Notation . . . 34
B.2 Displacement from initialization as a function of time . . . 34
B.3 Why norm-based uniform convergence is vacuous here . . . 36
B.4 PAC-Bayes centred at the initialization . . . 37
B.5 Distillation certificates without labelled hold-out data . . . 38
B.6 One-pass training: online-to-batch conversions . . . 39
B.7 Two-layer ReLU networks from balanced small initialization . . . 40
B.8 Quadratic networks with fixed output signs . . . 43

C Experimental details **45**

D How this paper was produced **46**

1 Introduction

For Bayesian neural networks there is a useful low-level picture of generalization: the posterior behaves like Occam learning over small circuits [15]. A network generalizes on a target when the target has a short description, and the sample cost is roughly that description length. This paper asks whether gradient descent admits a comparably concrete, discrete description, and how close it is to the Occam picture.

The literature already contains the qualitative ingredients. In the rich regime, from small initialization, gradient flow passes near a sequence of saddles. Each escape adds roughly one unit or one rank, so training resembles a greedy algorithm that grows a network one feature at a time. This has been made precise in several settings:

- diagonal linear networks [22];
- matrix factorization [20], deep linear networks [16] and deep ReLU networks [5];
- two-layer networks in general, through "alternating gradient flows" [18].

For sparse Boolean targets, the *leap complexity* of Abbe, Boix-Adserà and Misiakiewicz [2, 3] predicts that online SGD on a two-layer network needs on the order of $d^{\max(L-1,1)}$ steps. Here L is the largest number of new coordinates any Fourier monomial adds when the monomials are taken in the best order. This is the correlational-SQ (CSQ) rate [1, 6]. Evolution by mutation and selection is equivalent to CSQ learning [14, 23], and noisy neuroevolution is equivalent on average to gradient descent [24]. So the original idea behind this work, "a single circuit, mutated, keeping improvements", is already placed: it is CSQ learning.

What we add is quantitative and empirical. We train rich-regime networks on 41 Boolean targets (21 designed, 20 random formulas) at up to four input dimensions. We then build discrete models that reproduce the networks' sample complexity target by target, out of sample, and use ablations to find which ingredient of the model matters. We also prove a set of "training has not taken long, so the circuit is still small, so a union bound over small circuits applies" theorems, and test what can actually be certified for the trained networks.

Main findings.

1. **Gradient descent pays a search-noise cost, not a description-length cost** (Section 3). The networks' n^* grows like $d^{1.04}$, $d^{1.91}$ and $d^{2.40}$ for 2-, 3- and 4-bit parities (Table 2). An Occam learner needs about $k \log d$ samples.
2. **A greedy, noise-limited growth model matches the networks** (Section 4).
   - *The rule.* A candidate feature is accepted once $n\rho^2$ exceeds the noise level of the continuous search that would find it. Here ρ is its correlation with the residual, and the noise level grows by about a factor d/a for each new input beyond the second.
   - *Simulation.* The rule, simulated, matches n^* on 20 fresh random formulas within 1.15× on average.
   - *Closed form.* Evaluated in closed form along the target's greedy path, it predicted the 3-parity at $d = 120$ before the networks were run.
   - *Ablations.* The race dynamics we originally built into the model do not matter. The finite-sample noise of the direction search does.
3. **Memorize like a kernel, generalize like a feature learner** (Section 4.4). On partially learned cells, the networks' mistakes are best predicted by the network's own NNGP kernel.

A model that adds a kernel background to the grown features needs about 2 grown gates per such cell, where the pure growth model needs about 22.

4. **Bounds** (Section 6, Appendices A and B). The "short training ⇒ small" implication is a theorem for:

   - the discrete model;
   - diagonal linear networks;
   - quadratic networks with fixed output signs;
   - two-layer ReLU networks through the first escape.

   It is a conjecture beyond the first escape. On the real networks, the only tight bounds are a-posteriori distillation certificates.

5. **Composition** (Section 5). Leap complexity depends only on Fourier supports. We test whether intermediate features that are useful in their own right, and that compose into a higher-order term, make that term cheaper.

Scope. Everything empirical here is one architecture family, a μP two-hidden-layer ReLU MLP, mostly at width 128 and with Adam. Inputs are uniform on the hypercube, interactions have at most 4 bits, and the loss is square loss. We label every claim [Proved], [Conditional], [Conjecture], [Heuristic] or [Numerical].

2 Setup

Networks. The network is $f(x) = a^\top \mathrm{relu}(W_2\,\mathrm{relu}(W_1 x + b_1) + b_2)$ with two hidden layers of width $N = 128$ (32–256 in Section 3.4). It uses the μP (maximal-update) parameterization [25]:

- *Initialization:* $W_1 \sim \mathcal{N}(0, 1/d)$, $W_2 \sim \mathcal{N}(0, 1/N)$, a small readout $a \sim \mathcal{N}(0, 1/N^2)$, and zero biases.
- *Learning rates:* per-layer multipliers relative to a base width of 64.

Training minimizes the square loss on ± 1 labels with Adam [17], learning rate 10^{-3}. Minibatches of 64 are drawn with replacement from a fixed training set of size $n \in \{16, 32, \ldots, 16384\}$ (up to 65536 at $d = 60$) for 20000 steps, with 3 seeds (2 at $d \neq 30$). The weights move far from initialization, $\|\Delta W_1\|/\|W_1(0)\| \approx 0.7$–$1.05$, and the layer kernels' participation-ratio rank collapses from about 36 to 3–10. This is the feature-learning regime.

Two other protocols are used:

- *SGD:* μP-SGD with learning rate 0.2, on a subset of targets.
- *Online:* a fresh minibatch at every step, i.e. one pass over the data.

Targets. Inputs are uniform on $\{\pm 1\}^d$. There are 21 designed targets:

- a dictator x_1; AND_3; MAJ_3 and MAJ_9; a random halfspace on 10 bits;
- the parities $\chi_k = x_1 \cdots x_k$ for $k = 2, 3, 4$;
- a 5-term staircase $\mathrm{sign}(x_1 + x_1x_2 + \ldots)$;
- compositions: MAJ of parities, parity of majorities, recursive majority, $\mathrm{AND}(x_1x_2, x_3x_4)$;

- a multiplexer, a 3-term DNF, a decision list, $\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$;
- four random 6-gate formulas.

They are split into 12 *calibration* targets and 9 *held-out* targets.[1] A further 20 random fan-in-2 AND/OR/XOR formulas on 4–8 inputs (3–8 gates) were generated after all model design and calibration, as an out-of-sample test. Section 5 uses real-valued polynomial targets.

Metrics. We use the following:

- **Sample complexity n^*:** the smallest n at which the seed-mean test error at the end of training is at most 0.1, interpolated linearly in $\log n$ on the factor-2 grid.
- **Comparing a model with the networks:**
  - the mean $|\log_2(n^*_{\text{model}}/n^*_{\text{NN}})|$ over targets; 0.33 means within a factor of $2^{0.33} \approx 1.26$ on average;
  - the Spearman correlation of n^*;
  - *success agreement*, whether both have test error at most 0.1 at each (f, n);
  - the mean absolute difference of the test-error curves.
- **Per-example agreement:** computed on *partially learned cells*, where the networks' test error is in $[0.05, 0.45]$, with model and network trained on the same training set. It is the excess of their agreement on test points over the agreement that independent errors would give, as a fraction of the largest possible excess: 0 means independent errors, 1 means identical errors.
- **Time:** the Spearman correlation between a model's internal time to generalize and the networks' training steps to generalize.

Leap is the leap complexity of [3].

3 What the networks do

3.1 Sample complexity at $d = 30$

Table 1 and Figure 1 show the networks' learning curves against two reference learners:

- the NNGP kernel [19] of the same architecture, in kernel ridge regression;
- an Occam learner that picks the smallest junta, of up to 4 variables, consistent with the training set.

The networks sit between the two, in a specific way [Numerical]:

- **Sparse high-order targets.** Parities cost the networks 3–4× more data per extra bit: 200, 843 and 2728 samples. The Occam learner needs at most 16, 23 and 37, and the kernel fails from 3 bits on.
- **Dense leap-1 targets.** MAJ_9 and the 10-bit halfspace are 3–6× cheaper for the networks than for the kernel, and out of reach of the junta learner.

Several further observations [Numerical]:

[1] Calibration: x_1, MAJ_3, the halfspace, χ_2, χ_4, the staircase, parity of majorities, the DNF, recursive majority, $\mathrm{AND}(x_1x_2, x_3x_4)$, and two random formulas. Held-out: AND_3, MAJ_9, χ_3, MAJ of parities, the multiplexer, the decision list, $\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$, and two random formulas.

target	relevant	leap	n^* networks	n^* NNGP kernel	n^* Occam (juntas)
x_1	1	1	47	29	16
MAJ_3	3	1	112	73	18
MAJ_9	9	1	263	1470	–
random halfspace, 10 bits	10	1	563	1701	–
χ_2, χ_3, χ_4	2, 3, 4	2, 3, 4	200, 843, 2728	406, –, –	$\leq$16, 23, 37
staircase, 5 terms	5	1	1514	–	–
$\mathrm{MAJ}(x_1x_2, x_3x_4, x_5x_6)$	6	2	1380	1129	–
multiplexer $x_4\,?\,x_1x_2x_3 : \mathrm{MAJ}_3$	4	1	1156	–	37
3-term DNF	9	1	505	564	–
$\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$	6	3	2205	–	–

Table 1: Sample complexity at $d = 30$, width 128, Adam (selected targets). "–" means test error never reached 0.1 in the range tested ($n \leq 4096$ for the kernel, $n \leq 16384$ otherwise). The junta learner cannot represent targets with more than 4 relevant variables.

- **Reusing the training set saves data.** With fresh samples at every step, the networks need 12–64× their fixed-set n^*: about 12800 samples for χ_2, and about 32000 for χ_3, the staircase, the multiplexer and MAJ of parities.
- **Generalization is delayed near n^*.** At $n = 256$, χ_2 fits its training set by step 200 but generalizes only at steps 7000–10000. The multiplexer, the DNF, recursive majority and $\mathrm{AND}(x_1x_2, x_3x_4)$ show 4–40× delays at their critical n.
- **Features are learned in staircase order.** On the staircase target, the network's Fourier coefficient reaches half its target value at step 100 for x_1, step 500 for x_1x_2 and $x_1x_2x_3$, and step 1000 for $x_1x_2x_3x_4$.
- **Representation size peaks just below n^*.** For the 3- and 4-bit parities, the staircase and the multiplexer, the participation-ratio rank of the second layer's kernel peaks just below n^* and collapses above it. Below n^* the networks fit the training set with many features; above it they use the target's few features.

3.2 Dependence on the input dimension

Table 2 and Figure 2 show n^* at $d = 15, 30, 60$, and at $d = 120$ for four targets [Numerical]. The growth with d is polynomial, with an exponent that increases with the number of inputs that must be found at once:

- **Parities.** For 2- and 3-bit parities the exponents are 1.04 and 1.91 over $d = 15$–60. On a refined grid of n (quarter-octave steps), the local exponents from $d = 30$ to 120 are 1.09/1.03 and 1.93/1.96. Both match the CSQ rate $d^{\max(1,k-1)}$ [3, 6]. The 4-bit parity grows like $d^{2.40}$, below d^3.
- **Leap-1 compositions** (staircase, multiplexer) grow like $d^{1.5}$.
- **MAJ_9.** Its whole learning curve is a function of n/d at all four d, within 0.01–0.04 of a constant-free prediction from the sampling noise of the linear (Hebbian) direction (Figure 2b). Its apparent n^* exponent comes from where this flat curve crosses 0.1.

Occam-type laws grow like $\log d$, an effective exponent of about 0.2 over this range, and fail completely.

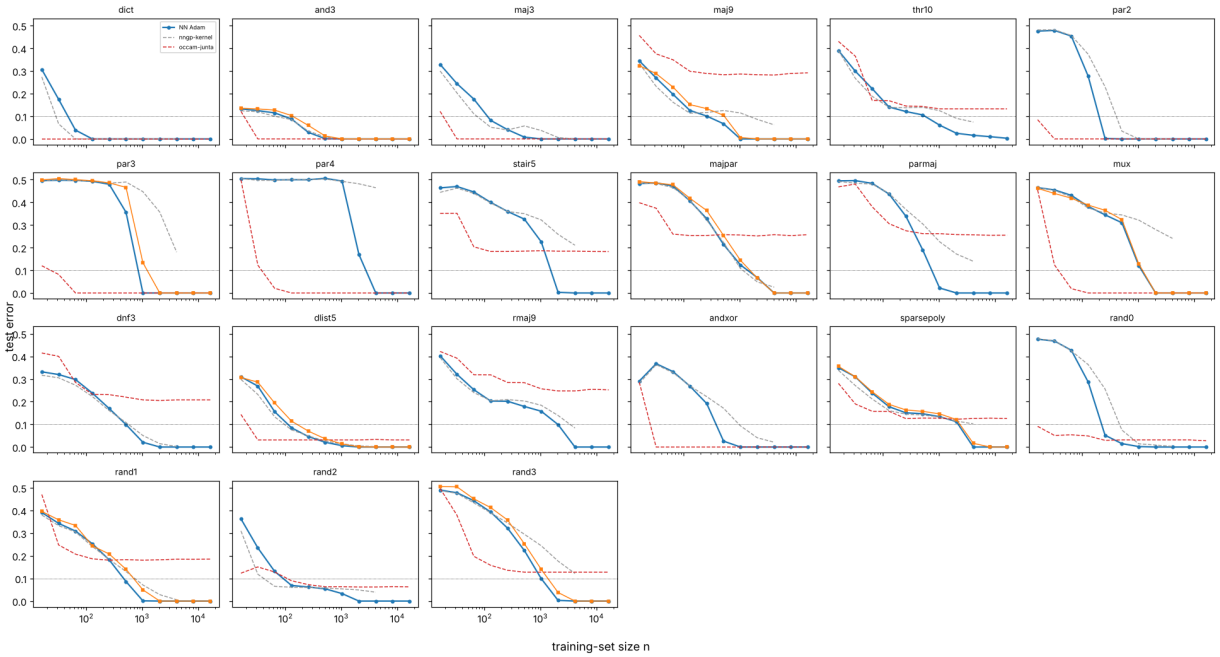


Figure 1: Test error at the end of training against training-set size, $d = 30$. Blue: Adam-trained networks (3 seeds). Orange: μP-SGD-trained networks, on the held-out targets (2 seeds). Grey dashed: NNGP kernel ridge regression. Red dashed: Occam over juntas of up to 4 variables.

Training reuses a fixed sample for many epochs, so the networks are not CSQ learners in the strict sense. SQ algorithms, or gradient descent that reuses batches, can reach $d^{k/2}$ in Gaussian single-index models [10, 11]. For $k \leq 3$ the networks here nevertheless follow the CSQ exponent; at $k = 4$ they fall between the two rates.

3.3 Gradient descent against Occam, by leap

Take the 26 targets, designed and random, with at most 4 relevant inputs. The ratio of the networks' n^* to the Occam-over-juntas learner's grows with the target's leap [Numerical]:

leap	1	2	3	4
targets	14	8	2	2
median $n^*_{\mathrm{NN}}/n^*_{\mathrm{Occam}}$ at $d = 30$	8.1	16.6	36	85

Within leap 1 the spread is wide, from 1.7× to 31× (the multiplexer is 31×). The correlations along the target's greedy path matter as well as its leap. Section 4.3 makes this quantitative.

3.4 Robustness: optimizer and width

SGD. μP-SGD on the same training sets gives n^* similar to Adam's on the held-out targets: mean $|\log_2|$ ratio 0.43 [Numerical]. On the 3-parity, SGD needs 382, 1220 and 3562 samples at $d = 15, 30, 60$, against Adam's 241, 843 and 3410. That is an exponent of 1.61 against 1.91. At $d = 60$ both values are interpolated inside the same factor-2 step of the n grid, so each exponent is uncertain by up to about ± 0.25, and the difference is not resolved.

target	n^* at $d = 15$ / 30 / 60 / 120	exponent (15–60)	CSQ d^{k-1}	SQ/reuse $d^{k/2}$
χ_2	100 / 200 / 427 / 869	1.04	1	1
χ_3	241 / 843 / 3410 / 10691†	1.91	2	1.5
χ_4	508 / 2728 / 14255 / –	2.40	3	2
staircase (5 terms)	433 / 1514 / 3487 / –	1.50	–	–
multiplexer	364 / 1156 / 2848 / 5195†	1.48	–	–
MAJ(x_1x_2, x_3x_4, x_5x_6)	306 / 1380 / 3466 / –	1.75	–	–
MAJ$_9$	123 / 263 / 625 / 845	1.17	–	–

Table 2: Network sample complexity against input dimension (Adam, width 128; 3 seeds at $d = 30$, 2 elsewhere). †On a refined (quarter-octave) n grid; the factor-2 grid gives 14188 and 6524. For χ_4 the 20000-step budget limits n^* at both ends of the range, so the sign of its effect on the exponent is unknown.

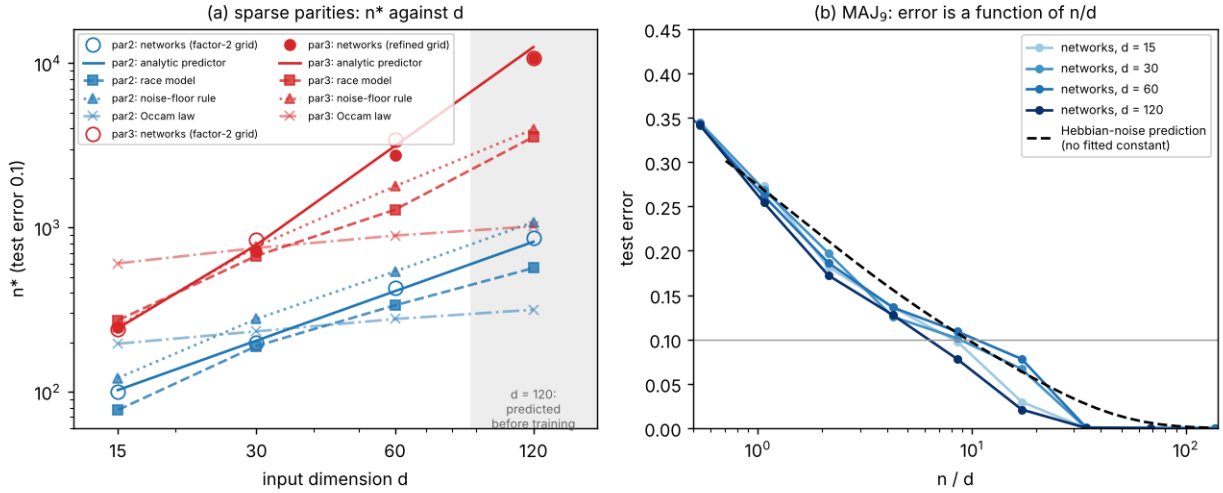


Figure 2: (a) n^* against d for the 2- and 3-bit parities: networks, the analytic noise-law predictor (Section 4.3), the race model, a noise-floor rule, and the Occam law. The predictions at $d = 120$ (shaded) were saved before the $d = 120$ networks were trained. (b) MAJ$_9$: network test error against n/d at four input dimensions, and the constant-free prediction from the noise of the Hebbian direction.

Width. At widths 32, 64, 128 and 256 (2 seeds; 3 at width 128) [Numerical]:

- **3-parity:** n^* does not change visibly (911, 827, 843, 873). At every width the test error is 0.28–0.47 at $n = 512$ and exactly 0 at $n = 1024$.
- **4-parity:** n^* falls monotonically with width (3554, 3304, 2728, 1899), a factor of 1.9 from width 32 to 256, and has not levelled off.

This pattern does not fit a single width-dependent constant in the noise law of Section 4.3, which would make the 4-parity's width effect the square of the 3-parity's. Nor does it fit a "luckiest unit" account. In that account the largest product of k initial weights over the units would set the escape, and it predicts a width effect for the 3-parity too, since that maximum grows 1.9× from width 32 to 256 at $k = 3$. A guess [Heuristic]: the 3-parity is at its width limit by width 32, its escape carried by the bulk of the units as in mean-field theory, while the 4-parity still relies on lucky units.

4 A discrete model of the feature channel

4.1 The race model

The first model, the *race model*, makes the saddle-to-saddle picture into an algorithm. Its parts:

- **State.** A shared two-level circuit of discrete ReLU gates $g = \mathrm{relu}(\sum_j w_j u_j - b)$. The weights are integers in $\{-3, \dots, 3\}$, and the wires u_j are input literals and earlier first-level gates. The gates feed a ridge least-squares readout, refitted after every new gate.
- **Proto-units.** A persistent pool of dormant proto-units of orders 1–4 stands in for the not-yet-recruited hidden units.
- **Alignment.** Each proto-unit's direction aligns to the residual by (tensor) power iteration on the training sample. It does so at the speed of the homogeneous law $\dot{m} = kU_k m^{k-1}$, with U_k the order-k utility.
- **Firing.** The rounded gate's utility then drives a clock, and the gate fires when the clock reaches $\theta = \log(1/\alpha) = 1.5$. The earliest gate to fire is added to the circuit, the readout is refitted, and the race resumes.
- **Stopping.** Growth stops at zero training error, or after 48 gates.

Three constants were calibrated on the 12 calibration targets only. The model sees nothing but the training sample.

Fit to the networks [Numerical]. Table 3 lists the scores.

- **Held-out targets.** The race model's n^* is within 1.26× of the networks' on average, it ranks the targets identically, and its learning curves match in shape. It reproduces intermediate plateaus. For example, on $\mathrm{sign}(2x_1 + x_2x_3 + x_4x_5x_6)$ test error stays near 1/8 from $n \approx 256$ to $n \approx 2000$, which is the error of the partial predictor $\mathrm{sign}(2x_1 + x_2x_3)$.
- **20 random formulas generated after calibration.** It is within 1.15× on average (Spearman 0.99), better than on the held-out set.
- **Other input dimensions.** At $d = 15$ and $d = 60$, without recalibration, it is within about 1.35×.
- **The original "mutate and keep improvements" model.** A strong-selection/weak-mutation evolution of a single fan-in-2 circuit does much worse: mean $|\log_2|$ ratio 1.36.

4.2 What makes it work: ablations

Table 4 summarizes 30 ablations, each run on the full grid with the calibrated constants and no recalibration [Numerical]. Every "no change" is judged against the model's own variation across its internal random streams, which spans 0.23–0.33 held-out over six streams.

- **The race dynamics do not set n^*.** Removing the alignment cost, resetting the clocks after every firing, or replacing the race by plain greedy selection changes held-out n^* error by at most about 0.1 beyond that spread. So does varying θ from 0.5 to 6, the power-iteration steps from 1 to 32, or the pool size from 1 to 32. Compared model against model, these variants differ from the calibrated model by 0.13–0.21 in $|\log_2 n^*|$, no more than two random streams differ from each other (0.11–0.23).

model	mean $\lvert\log_2(n^*_{\text{model}}/n^*_{\text{NN}})\rvert$ held-out (9)	random (20)	$d = 15$	$d = 60$	per-example	time Sp.
race model	0.33	**0.20**	0.45	0.43	0.50	0.83
race + kernel background (Section 4.4)	0.35	–	–	–	**0.65**	–
minimal model (2 constants)	0.59	0.47	0.30	0.52	0.50	0.69
analytic predictor (no simulation)	0.38	0.42	0.30	0.52	–	–
sswm (mutate and select)	1.36	1.12	–	–	0.26	0.72
NNGP kernel	2.58	1.78	–	–	0.69	0.22
Occam over juntas	4.61	3.76	–	–	0.20	−0.07
SGD-trained vs Adam-trained networks	*0.43*	–	–	–	*0.71*	*0.83*
Adam networks from a new initialization	–	–	–	–	*0.62*	–

Table 3: Models scored against the Adam-trained networks. Per-example agreement and the time Spearman are on the held-out targets. The race model's held-out score varies between 0.23 and 0.33 across its internal random streams; its value averaged over streams is 0.28. The hybrid's score is the average over its streams. The $d = 15$ agreement of the minimal model and the predictor depends on a cap that was probably added after the $d = 15$ network results had been seen, so it is not a clean out-of-sample test. The last two rows compare networks with networks.

- **The noise of the direction search does.** Computing the search directions on a large fresh sample removes their finite-sample noise. n^* then collapses to 16–87 on every target, as far from the networks as the Occam learner (4.64). Computing them on an independent sample of the *same* size n restores the match (0.31). So what matters is the noise *level* at sample size n, not overfitting to the particular training labels.
- **The particular sample's noise sets which examples are wrong.** With the independent search sample, per-example agreement falls from 0.50 to 0.26, although n^* is matched. Roughly half of the race model's excess agreement with the networks comes from both being misled by the same spurious correlations of the shared sample [Heuristic].
- **The gate library matters.** Without order-3 and order-4 candidates, the 3- and 4-bit parities, the sparse polynomial and recursive majority are never learned (1.43). Threshold gates in place of ReLU gates score 0.60, and weights in $\{-1, 0, 1\}$ score 0.81. Depth-2 gates help only staircase-like targets, by about 0.4 in $\log_2$.
- **Time.** The ordering of targets by time to generalize survives every ablation, including the noise-free search, so the true features' utilities set it, not the race clocks. Within a target, the time Spearman is about 0.

4.3 The noise law and a closed-form predictor

The ablations suggest a smaller model [Heuristic]: greedy stagewise least squares over ReLU gates with ± 1 weights, in which a candidate is accepted when

$$n\rho^2 \;>\; c\,N(k), \qquad N(k) = d\,\max\!\big(1, \tfrac{d}{a}\big)^{\max(k-2,0)}\ (k \geq 1), \qquad N(0) = |\text{active pool}|. \tag{1}$$

- ρ is the gate's correlation with the current residual.
- k is the number of inputs the gate needs that the circuit does not yet use. Reusing grown features ($k = 0$) costs only a search over the small active pool.

variant	held-out $\lvert\log_2\rvert$ ratio	per-example
race model (calibrated)	0.33	0.50
greedy: add the highest-utility gate, no race	0.27	0.53
one alignment law for all orders	0.28	0.50
depth-1 gates only	0.35	0.51
two other internal random streams	0.32, 0.27	0.51, 0.51
noise-free search (directions from a large fresh sample)	4.64	0.04
independent search sample of the same size n	0.31	0.26
no order-3/4 candidates	1.43	0.52

Table 4: Selected ablations of the race model, held-out targets.

- The heuristic derivation comes from the alignment law. An order-k unit escapes only if its signal $kUm(0)^{k-1}$ beats the frozen sampling noise $\sim n^{-1/2}$, with an effective initial overlap $m(0)^2 = a/d$.

This is the correlational-SQ / leap scaling $d^{\max(1,k-1)}$. Two constants, $c = 2.25$ and $a = 18$, were calibrated on the calibration targets.

Evaluated along the target's *population* greedy path, (1) gives a closed form:

$$n^* \approx \max\Big(n_{\text{Hebb}},\ c \max_{i \le i^*} \frac{N(k_i)}{\rho_i^2}\Big).$$

Here i^* is the shortest prefix of the path that reaches error 0.1, and n_{Hebb} is the sampling-noise floor of the linear part. Some closed forms:

- $n^*(x_1) = \Phi^{-1}(0.9)^2 d = 1.64\, d$, with no fitted constant.
- $n^*(\text{MAJ}_9) = 9.52\, d$, with no fitted constant.
- $n^*(\chi_k) = c\,(2^k - 1)\, d \max(1, d/a)^{k-2}$. The first gate, AND_k, has $\rho^2 = 1/(2^k - 1)$ with χ_k, and the remaining gates are cheap $k = 0$ moves.

On the 20 random formulas the closed form is within 1.34× on average (Spearman 0.92–0.96, depending on the saved run). The simulated minimal model keeps the race model's success agreement and per-example agreement, but not its held-out n^* accuracy (0.59, against an average of 0.28 for the race across its streams).

A pre-registered test at $d = 120$ [Numerical]. The predictor, its variants, and the race model (4 internal streams) were evaluated at $d = 120$ with their $d = 30$ constants. Their predictions were saved before the $d = 120$ networks were launched, and the file timestamps confirm the order.

target	networks	analytic predictor	race model	Occam law
χ_2	869	**817**	566	315
χ_3	10691 (refined grid)	**12525**	3557	1015
MAJ_9	845	1143	1307	1143
multiplexer	5195 (refined grid)	3405	4013	891

The 3-parity is the decisive case.

- **Predictor.** The leap law predicted that n^* would roughly quadruple from $d = 60$ to 120. It did: from 2740 to 10691 on the refined grid.

- **Race model.** It grows only like $d^{1.2}$ and is now 3× too optimistic. Its order-k search accepts a direction once it is *detectable* in the empirical moments. The networks behave as if the direction must be *reached* from a random start.
- **Why the race model matched at $d = 30$.** At $d = 30$ the two thresholds nearly coincide [Heuristic].
- **Misses of the predictor.** It underpredicts compositions of low-leap moves: the multiplexer by 1.5× at $d = 120$, and by 1.4–2.1× in general. It overpredicts the 4-parity at $d = 60$ by 1.6×.

4.4 Kernel background plus grown features

The race model matches *whether* and *when* the networks generalize, but agrees less with *which* test points they get wrong (0.50) than the NNGP kernel does (0.69). A diagnosis on the 95 partially learned cells [Numerical]:

- **The output has two parts.** Split each learner's output into the part that depends only on the target's relevant variables, the signal, and the rest, which comes from the particular training sample. About 45% of a network's output variance is this sample noise: 59% at $n \leq 64$, 23% at $n = 2048$.
- **Every learner gets the signal; the kernel gets the noise.** All learners match the network's signal part (correlation 0.92–0.98). The noise part correlates 0.81 with the kernel's, 0.70 with that of a network retrained from a new initialization, and 0.48 with the race model's.
- **Swapping only the noise patterns** moves the race model's agreement from 0.50 to 0.63, and the kernel's from 0.69 to 0.53.
- **Over training time,** the networks first interpolate within 100–1000 steps, with kernel-like mistakes. Near n^* the generalizing features grow afterwards.
- **The kernel behaves like the network averaged over initializations.** Two single networks trained on the same data from different initializations agree only at 0.62. The average of three such networks agrees with a fourth at 0.684, and this extrapolates to about 0.71–0.73. The kernel agrees 0.78 with a three-network average.

A hybrid model adds an NNGP-kernel background to the race's gates in one joint ridge readout [Numerical]:

- **The background** relaxes on the race clock like kernel gradient flow, with ridge $c_{\mathrm{bg}} n/t$.
- **Growth** stops at a square-loss level, `stop_mse`, after which the background interpolates the remaining residual.
- **Calibration.** Two constants, c_{bg} and `stop_mse`, were chosen on the calibration set, together with the schedule form, the stop rule and the end-of-training step.

Its results, and what they suggest:

- **Agreement.** On the held-out targets it raises per-example agreement from 0.50 to 0.65 (bootstrap over targets: +0.15, interval $[0.13, 0.17]$). The gain holds on every held-out target and every seed.
- **Cost.** Its n^* error is 0.35 against the race model's 0.28, both averaged over random streams.

- **Few grown gates.** It needs about 2 grown gates per partially learned cell, where the race model needs about 22. The race model's extra gates memorize the sample. The networks do that with their random-feature background instead [Heuristic].
- **A candidate bound.** This suggests an a-priori bound of the form "Occam for a few grown gates, plus a kernel-norm term for the background". We have not attempted it.
- **Remaining discrepancy.** On easy targets the networks are 1.1–2.2× *less* sample-efficient than their own NNGP kernel, for example 47 against 29 samples for x_1. The race model reproduces this through its noisy Hebbian first gate; the hybrid does not.

5 Does compositional structure beyond leap matter?

(Results pending.)

6 Generalization bounds

The question here is whether "training has not taken long, so the circuit cannot have grown big, so a Chernoff bound over all small circuits applies" can be made rigorous, and whether it gives useful numbers. Appendices A and B state the results precisely. This section summarizes them.

6.1 What is proved

The Occam step needs no dynamics [Proved]. Consider any learner whose output after T gates is a circuit from a fixed library, followed by a threshold readout. In the realizable case its test error is at most about

$$(T(\log|G| + k\log(d+T)) + T^2\log 2 + \log\tfrac{(T+1)(T+2)}{\delta})/n,$$

simultaneously for all T (Appendix A). The dynamics are needed only to bound T.

Training time does not bound the raw gate count. Many correlated gates can fire at once. On MAJ_m, m^2 gates fire at the same instant of the race. So training time cannot bound the raw gate count uniformly over gate libraries. For a fixed library only a weak time-only bound holds, with a factor of about d. What time does bound, in the race model, are three quantities [Proved]. Time is measured as $\tau(t) = \int_0^t \sqrt{2\hat{L}}$ in units of $\Theta = \log(1/\alpha)$, and the three quantities are:

- the circuit's depth;
- its incoherence-weighted size, $T \le \kappa(\tau/\Theta)^2$, where κ is the top eigenvalue of the fired gates' Gram matrix;
- the readout's ℓ_1 norm, at depth one.

Each gives a test-error bound as a function of training time. The race reaches the noise floor only at $t \gtrsim \Theta\sqrt{n/\log M}$. This is a precise version of "it has not had time to memorize yet".

Diagonal linear networks [Proved]. For a diagonal linear network over a Boolean feature dictionary, small-initialization gradient flow keeps the number of large coordinates at most $\kappa\tau^2/\Theta_\varepsilon^2$. This gives a generalization bound uniform in time, with only ln(dictionary size) dependence.

Two-layer quadratic networks with fixed output signs [Proved]. These are networks $f(x) = \sum_j[(u_j \cdot x)^2 - \|u_j\|^2] - \sum_j[(v_j \cdot x)^2 - \|v_j\|^2] = \langle UU^\top - VV^\top, xx^\top - I\rangle$, with output weights fixed at ± 1 and centred units, equivalently symmetric matrix sensing. Under gradient flow from small initialization, the number of grown hidden units is at most $\kappa(\tau/\Theta)^2$ at every training time.

- κ depends only on the training inputs: not on the labels, the trajectory or the width.
- The bound is weight-free rather than a-priori: τ is read off the training-loss curve. The same statement in raw training time alone is much weaker.

Together with a union bound over low-rank classes, this gives a test-error bound uniform in time. This makes the motivating sketch rigorous, for this class of networks.

Two-layer ReLU networks (gradient flow from balanced small initialization). We prove three things [Proved]:

- an exact "mass clock" identity;
- that no unit grows before about $\log(1/\alpha)/U^*$, with constant 1, which is sharp;
- a width-free generalization bound valid through the first escape from the initial saddle. It is vacuous below about $n \approx 10^5$ at $d = 16$.

In addition:

- **Merging.** A conserved quantity, the off-diagonal part of the classical balancedness law [4, 12], makes same-sign neurons that share an activation pattern on the training set merge as their mass grows.
- **Beyond the first escape.** Persistence of a small circuit holds under a condition that can be checked from the training residuals [Conditional]. The a-priori statement is open [Conjecture].
- **Reservoir depletion.** In small-α simulations every neuron was recruited by the first escape. In one run at $\alpha = 10^{-6}$ the second feature was then not learned by $t = 400$. The discrete models assume the reservoir of dormant units persists; whether it does in the μP networks was not measured.

One-pass training [Proved]. With fresh minibatches, the prequential error certifies the average test error of a window of iterates, with no capacity argument [9]. Taking the window in advance with N fresh samples and M prequential mistakes, the bound is $M/N + \sqrt{\ln(1/\delta)/(2N)}$.

6.2 What can be certified for the trained networks

A-priori bounds are vacuous [Numerical]. Norm-based, spectral, stability and mutual-information bounds are vacuous at every $n \le 16384$, even at initialization. PAC-Bayes with a prior centred at initialization [13, 21] reaches 0.26–0.56 at $n = 16384$ on the simplest targets only.

Distillation certificates are non-vacuous and never violated [Proved method; Numerical results]. The method:

- Find a small circuit g that agrees with the trained network on *unlabelled* inputs. The candidates are a junta truth table, or the sign of a few Fourier characters with small integer weights.

- Bound the network's test error by an Occam/Chernoff bound for g on the training set, plus the measured disagreement.

No labelled hold-out data is used. Over all 21 targets, 5 training-set sizes and 3 seeds:

- 221 of 315 networks get a certificate below 0.5, and 163 one below 0.2.
- No certificate falls below the network's actual test error.
- At $n = 16384$, every network is certified at 0.08 or less, except on three targets, where the certificate's circuit classes do not capture the network cheaply: MAJ of pair parities (0.54), sparse polynomial (0.29) and one random formula (0.42).
- Over training time, the certificates track delayed generalization. On χ_3 at $n = 1024$, the network fits its training set by step 1000 with test error 0.165, and the certificate is 0.23. By step 20000 the test error is 0 and the certificate 0.073. While the network memorizes, the best certifying class temporarily grows, for example to signs of 8–10 characters, and then collapses to the 3-bit parity.

The race model's own circuits. When the race model's circuit is small, its own-circuit Occam bound is tight: median 0.021 among generalizing fits with at most 6 gates. Half of its generalizing fits carry more than 6 gates, and those bounds are vacuous (median 0.67). Yet most of these large circuits are *functionally* small. A distillation certificate has median 0.125 on a sample of 30 and is below 0.2 in 24 of them. Size of representation and size of function come apart, for the model and for the networks alike.

Online. On the one-pass runs, the prequential bound is within 0.01–0.18 of the test error at the end of each window throughout learning, and about 0.002 after learning. On MAJ_9 it is 0.53, 0.37, 0.20, 0.074, 0.032 and 0.013 over successive windows.

6.3 What the bounds do not answer

The bounds control the gap between training and test error. In every case here, once gradient descent generalizes, the Occam bound for the circuit it found is already small: the target's circuit is tiny and $n \geq n^* \gg k \log d$. The binding question is therefore not the gap but *how many samples until the feature is found*, i.e. the noise law of Section 4.3:

- **Lower half.** For one-pass (CSQ-type) training, its lower half is the known CSQ rate [1, 3]. For training that reuses a fixed sample, no matching lower bound is known, and the 4-parity data scale below d^{k-1}.
- **Upper half.** For real networks, its upper half is open.

7 Discussion

7.1 The picture

Our current picture of rich-regime gradient descent on these targets [Heuristic], summarizing the evidence of Sections 3 and 4:

$$\text{trained network} \approx \underbrace{\text{random-feature kernel background}}_{\text{fits the sample noise; decides which examples are wrong}} + \underbrace{\text{a few grown features}}_{\text{decide whether and when it generalizes}} .$$

Features are added greedily. A candidate is accepted once its correlation with the current residual beats the sampling noise of the *continuous* search that would find it. That noise depends on how many inputs the feature needs at once that the circuit does not yet use, and it grows polynomially in d. It is not the best-of-M noise of a union bound over a finite library, so the cost is not a description length.

7.2 Relation to NNBayes ≈ Occam

- **The same answer to *what* is learned.** Once gradient descent generalizes, the network is functionally close to a small circuit, as the distillation certificates show. So does the Occam picture predict [15].
- **Different answers to *how many samples*.**
  - Occam, and the $\gamma \geq 1$ posterior of [15], pays the description length, about $k \log d$ for a k-parity.
  - Gradient descent pays the noise cost of the most expensive move along its greedy path, about d^{k-1} for a move that needs k new inputs at once.
- **The gap grows with leap:** the median ratio is 8×, 17×, 36× and 85× at leaps 1–4, at $d = 30$.

A conjecture suggested by the predictor, not shown here [Conjecture]: the two sample complexities are polynomially related exactly for targets with a cheap greedy path, i.e. low-leap compositions with large residual correlations at every step. They differ by about $d^{k-1}/(k \log d)$ when a leap of size k is unavoidable.

7.3 What is in the literature, and what is new

Known. The qualitative picture is largely in the literature:

- the saddle-to-saddle / incremental-learning view of rich-regime gradient descent [5, 16, 18, 20, 22];
- the leap / CSQ account of which sparse targets online SGD learns, and how fast [2, 3, 6];
- the equivalence of mutation-and-selection with CSQ learning [14].

New here, as far as we know. The contributions are quantitative and empirical:

1. a calibrated discrete model and a two-constant closed form that predict per-target sample complexity of multi-epoch, fixed-sample training out of sample, including a pre-registered prediction at $d = 120$;
2. ablations showing that the finite-sample noise level of the direction search, not the race dynamics, is the operative ingredient;
3. the kernel-background decomposition of the networks' mistakes;
4. the specific theorems of Appendices A and B: the weight-free rank bound for quadratic networks, and the early-phase results for ReLU networks;
5. the empirical finding that distillation certificates are the only tight bounds in this setting.

7.4 Leap counts supports, not structure

Leap complexity is a function of the target's Fourier *supports* only. Once the inputs of a monomial have been found, any further function of them is free in d. Leap therefore cannot see whether a high-order term can be built from intermediate features that are useful in their own right. Our move classification in (1) inherits part of this blindness, since k counts new inputs. But the correlation ρ of the best available gate depends on which features already exist, so the noise law can in principle see such structure. An early version of this work proposed a "circuit leap" complexity. It orders a circuit's gates so that each insertion is a positive-utility move, and counts the largest number of gates that must be inserted at once. Section 5 tests the question directly.

7.5 Limitations

- **One setting.** All experiments use one architecture family (a μP two-hidden-layer ReLU MLP), mostly width 128 and Adam, square loss, uniform Boolean inputs, $d \leq 120$, and interactions of at most 4 bits.
- **Width.** The 4-parity's width dependence is not explained by any model here (Section 3.4).
- **The noise law.** Its form for 3 or more simultaneous inputs is a fit, not a derivation. It misses compositions by up to 2.1×, and it overpredicts the 4-parity at $d = 60$.
- **Training budget.** The networks' n^* is still falling at 20000 steps, by about 0.17 in $\log_2$ per doubling of training. Part of every match is a match at this training budget.
- **Verification.** The theorems were checked by AI referees, not by humans (Appendix D).

7.6 Open problems

1. **Prove the noise law** for a two-layer network trained on a fixed sample from small initialization: an order-k feature escapes the plateau iff $n \gtrsim d^{k-1}/\rho^2$, up to logarithms. Ingredients exist: information-exponent results for online SGD [6], early alignment [8], and the mass clock and merging law of Appendix B.
2. **Composition costs, and width dependence at order 4.**
3. **Reservoir depletion.** When does the first escape use up the dormant units at practical scales?
4. **A bound of the form "Occam for a few grown features plus a kernel norm for the background"**, as suggested by the hybrid model.
5. **Bayes versus gradient descent.** Characterize the targets on which their sample complexities are polynomially related.

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A Bounds for the discrete circuit-growth model and for diagonal linear networks

This appendix states the results of two derivations. The first is `theory/A-discrete-model.md`, Sections A1–A4, on the discrete circuit-growth model. The second is Sections C2–C3 of `theory/C-bridge-and-certif` on diagonal linear networks and on a conditional transfer to ReLU networks. Each derivation was refereed twice, and the version stated here is the one after both revisions. The referee reports and check scripts are in `theory/checks`. Numbering in parentheses is the derivation's. Proofs are only sketched; the derivations give full proofs with explicit constants. A result marked "Literature, from memory" is a published result whose statement the derivation could not check against the paper. No result here has been reviewed by a human.

A.1 Setting, notation and counting

Data. $\mathcal{X} = \{\pm 1\}^d$, and P is any distribution on $\mathcal{X} \times \{\pm 1\}$; the experiments use uniform x and $y = f^*(x)$. The training sample $S = ((x_i, y_i))_{i \le n}$ is iid from P. We write $\langle u, v \rangle_n = \frac{1}{n} \sum_i u(x_i, y_i) v(x_i, y_i)$, $\| \cdot \|_n$ for its norm, $\bar{u} = \langle u, 1 \rangle_n$, $\mathrm{err}(h) = P(h(x) \neq y)$, and $\widehat{\mathrm{err}}(h)$ for its empirical version on S. Logarithms are natural. $\mathrm{kl}(q \| p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p}$ is the binary relative entropy, and $\mathrm{kl}_+^{-1}(q, \epsilon) = \sup\{p \in [q, 1] : \mathrm{kl}(q \| p) \le \epsilon\}$. It satisfies $\mathrm{kl}_+^{-1}(q, \epsilon) \le q + \sqrt{2q\epsilon} + 2\epsilon$ and $\mathrm{kl}_+^{-1}(0, \epsilon) = 1 - e^{-\epsilon} \le \epsilon$. A bound on a test error is *useful* if it is at most 0.1, and *non-vacuous* if it is below $1/2$.

Circuits. A *library* G is a finite set of gate functions $\phi : [-1,1]^j \to [-1,1]$ of arity $j \le k$. It is *Boolean* if every ϕ maps $\{\pm 1\}^j$ into $\{\pm 1\}$. A *circuit with T gates* is a list $(\phi_t, \iota_t)_{t \le T}$ with $\phi_t \in G$ and source indices $\iota_t \in \{1, \dots, d+t-1\}^{j_t}$, where indices $1, \dots, d$ are inputs and $d+s$ is gate s; gate t computes $g_t = \phi_t(\text{sources})$. A *predictor* is $h = \psi(g_1, \dots, g_T)$, with a *sign readout* $\psi(z) = \operatorname{sgn}(c_0 + c \cdot z)$, an *output-gate readout* $\psi(z) = \pm z_T$, or a real readout $c_0 + c \cdot z$. $\mathcal{H}_T$ is the set of predictors on circuits with T gates. A *learner* is any rule, possibly randomized and arbitrarily data-dependent, that maps S to a sequence of circuits and readouts. The running example is the *toy library* $\{\mathrm{id}, \mathrm{AND}(\pm a, \pm b), \mathrm{XOR}(a,b)\}$ up to negation, with $|G| = 6$ and $k = 2$. At $d = 30$ it has 2205 depth-one candidates.

Lemma A.1 (Counting (Lemmas 1.1 and 1.2 in the derivation))**.** *[Proved] There are at most e^{W_T} circuits with T gates, where $W_T := \sum_{t=1}^{T} (\log|G| + k \log(d+t-1)) \le T(\log|G| + k\log(d+T-1))$. On a set $Z \subset \mathbb{R}^T$ of N points, the number of functions $z \mapsto \operatorname{sgn}(c_0 + c \cdot z)$ (with either fixed convention for $\operatorname{sgn}(0)$) is at most $\Phi(N,T) := 2\sum_{i=0}^{T} \binom{N-1}{i} \le 2N^T$. Hence $\log|\mathcal{H}_T| \le W_T + \rho_T$, where $\rho_T := \log \Phi(2^{\min(T,d)}, T)$ for a Boolean library (about $T^2 \log 2$ for $T \le d$), $\rho_T := \log \Phi(2^d, T)$ for real-valued gates, and $\rho_T := \log 2$ for the output-gate readout.*

The second count is the upper half of Cover's (1965) function-counting theorem. For $T \le 4$ and $Z = \{\pm 1\}^T$, linear programming gives the exact counts $4, 14, 104, 1882$, against $\Phi = 4, 14, 128, 3882$ [Numerical].

A.2 The Occam bound, uniform over circuit size

Theorem A.2 (Occam bound, uniform over circuit size (Theorem A1 in the derivation))**.** *[Proved] Fix $\delta \in (0,1)$ and put $L_T := \log|\mathcal{H}_T| + \log\frac{(T+1)(T+2)}{\delta} \le W_T + \rho_T + \log\frac{(T+1)(T+2)}{\delta}$. With probability at least $1-\delta$ over S, the following hold simultaneously for every $T \ge 0$ and every $h \in \mathcal{H}_T$, where $p = \operatorname{err}(h)$ and $\hat{p} = \widehat{\operatorname{err}}(h)$:*

(1) if $\hat{p} \le p$, then $\mathrm{kl}(\hat{p}\|p) \le L_T/n$;

(2) $p \le \hat{p} + \sqrt{2\hat{p}L_T/n} + 2L_T/n$ (Bernstein form);

(3) $p \le \hat{p} + \sqrt{L_T/(2n)}$ (agnostic form);

(4) if $\hat{p} = 0$, then $p \le 1 - e^{-L_T/n} \le L_T/n$ (realizable form).

A learner's output after $T(S)$ gates lies in $\mathcal{H}_{T(S)}$, however the gates were chosen. So the four bounds hold for the learner at every stage and at every training time, with T the current number of distinct gates in use. Gates that compute the same function, or are unused, can be dropped without changing h.

Proof sketch. For one fixed h, the Chernoff bound in kl form gives $P(\hat{p} \le p,\ \mathrm{kl}(\hat{p}\|p) > \epsilon) \le e^{-n\epsilon}$. Take $\epsilon = L_T/n$, and take a union bound over $\mathcal{H}_T$ and then over T; the weights $\delta/((T+1)(T+2))$ sum to δ. Items (2)–(4) are readings of (1), through $\mathrm{kl}(q\|p) \ge (p-q)^2/(2p)$ for $q \le p$, Pinsker's inequality, and $\mathrm{kl}(0\|p) = -\log(1-p)$. Nothing about the learning dynamics is used. This is the classical Occam argument [7], made uniform over T. □

Values [Numerical]. For the toy library at $d = 30$ and $\delta = 0.05$, $L_T = 14.8, 151, 427$ nats at $T = 1, 10, 20$. The realizable bound (4) is useful for $T \le 2, 7, 20, 62$ and non-vacuous for $T \le 11, 27, 102, 403$, at $n = 256, 1024, 4096, 16384$ respectively. For $1 \le T \le 10$, $n \ge 144\,T$ suffices for a useful bound. The agnostic form (3) is much weaker: at $T = 5$ its gap is 0.35 at $n = 256$ and 0.088 at $n = 4096$. A symmetrization route in the style of Blumer, Ehrenfeucht, Haussler and

Warmuth (1989) replaces ρ_T by about $T\log(2n)$, but it multiplies the whole complexity by $2/\ln 2$. With these constants it is larger than (4) at every $T \le 1499$ for $n \in \{1024, 4096, 16384\}$; at $T = 10$ and $n = 4096$, for example, it gives 0.122 against 0.036.

Theorem A.3 (Real readouts with an ℓ_1 constraint (Theorem A1′ in the derivation))**.** *[Proved, given three cited textbook lemmas] Let $\mathcal{F}$ be a finite set of functions $\mathcal{X} \to [-1,1]$, for example the depth-one gates, with $|\mathcal{F}_1| \le |G|d^k$. Let $\mathcal{A}_B = \{c_0 + \sum_j c_j g_j : g_j \in \mathcal{F},\ |c_0| \le 1 + B,\ \sum_j |c_j| \le B\}$. Let $\ell_{\mathrm{sq}}(f) = \frac{1}{2}(\mathrm{clip}_{[-1,1]} f - y)^2$, let $\varphi_\theta(u) = \min\{1, \max\{0, 1 - u/\theta\}\}$ be the ramp loss, and put $B_j = 2^j$, $\pi_j = 1/((j+1)(j+2))$ and $\mathfrak{R}_n(B) = \frac{1+B}{\sqrt{n}} + B\sqrt{2\log(2|\mathcal{F}|)/n}$. Fix θ in advance. With probability at least $1 - \delta$, for all $j \ge 0$ and all $f \in \mathcal{A}_{B_j}$,*

$$\mathbb{E}\ell_{\mathrm{sq}}(f) \le \hat{\mathbb{E}}\ell_{\mathrm{sq}}(f) + 4\mathfrak{R}_n(B_j) + 2\sqrt{\tfrac{\log(2/(\pi_j\delta))}{2n}}, \qquad \mathrm{err}(\mathrm{sgn} f) \le \hat{\mathbb{E}}\varphi_\theta(yf) + \tfrac{2}{\theta}\mathfrak{R}_n(B_j) + \sqrt{\tfrac{\log(2/(\pi_j\delta))}{2n}}.$$

A predictor with $\sum_j |c_j| = B$ and $|c_0| \le 1 + B$ lies in some $\mathcal{A}_{B_j}$ with $B_j \le 2\max(B, 1)$.

The three cited lemmas are the Rademacher uniform-deviation bound, Talagrand's contraction lemma and Massart's finite-class lemma, in the forms of Mohri, Rostamizadeh and Talwalkar (2018). The bound does not depend on the number of gates, but its constants are poor (Appendix A.5).

A.3 The clock race

Definition A.4 (Clock race (Definition 2.1 in the derivation))**.** Fix a threshold $\Theta > 0$. At time $t \ge 0$ the state is a set $A(t)$ of *fired* gates, which forms a circuit; a readout $c(t)$; and the training residual $r(t) = y - f_t$, where $f_t = c_0(t) + \sum_{m \in A(t)} c_m(t) g_m$.

- A *candidate* m is a gate (ϕ, ι), $\phi \in G$, whose sources are inputs or fired gates. Its *availability time* a_m is the latest firing time among its gate sources, or 0 if it reads only inputs.
- Its *utility* is $U_m(t) = \langle r(t), g_m \rangle_n$, and its *clock* is $S_m(t) = \int_{a_m}^t U_m(s)\, ds$.
- A candidate fires the first time $|S_m(t)| = \Theta$, with orientation $\sigma_m = \mathrm{sgn}\, S_m$. Any number of candidates may fire at once, and clocks are never reset.
- After each event the readout is refit, in one of two variants:

  (LS) least squares with a free intercept over all fired gates, at any depth;

  (SC) for a depth-one dictionary only: least squares with a free intercept over the *saturated* set $\mathrm{Sat}(t) = \{m : |S_m(t)| = \Theta\}$, subject to $\sigma_m c_m \ge 0$, with $c_m = 0$ off it.

The *effective time* is $\tau(t) = \int_0^t \|r(s)\|_n\, ds = \int_0^t \sqrt{2\hat{L}(s)}\, ds$, where $\hat{L}$ is the training square loss. The *integrated residual* is $R(t) = \int_0^t r(s)\, ds \in \mathbb{R}^n$. The *population race* replaces $\langle \cdot, \cdot \rangle_n$ by $L^2(P)$ inner products.

Basic facts (Lemma 2.2 in the derivation) [Proved].

- $|U_m| \le \|r\|_n \le 1$, so $\tau(t) \le t$.
- In (LS), a fired gate has $U_m = 0$ afterwards (by the normal equations), so its clock stays at $\sigma_m\Theta$. In (SC), $|S_m| \le \Theta$ always.
- If all candidates share one threshold, multiplying Θ by λ multiplies every event time by λ.

Every (SC) statement below is proved for all t before the first accumulation point of event times, if there is one. Finiteness of (SC) events is open, but all numerical (SC) runs had finitely many.

What the race models.

- *Time unit.* For moves that each need $k = 2$ fresh weight layers, $\Theta = \log(1/\alpha)$ up to a parametrization constant, and by scale invariance α only sets the unit of time. This fails once *fresh chains* ($k \geq 3$ new weight layers in series) are included, since their thresholds grow like $\alpha^{-(k-2)}$. In a three-layer network started entirely at scale α, the first move is such a chain and costs time about α^{-1} [Heuristic]. The depth-one race therefore models diagonal or 2-homogeneous networks, not the first move of a three-layer network.
- *Depth one.* For diagonal linear networks, the small-α limit with time rescaled by $\log(1/\alpha)$ is a saddle-to-saddle process [22]. As far as the derivation recalls, that process corresponds to (SC), except that the diagonal network has no free intercept (Literature, from memory). (LS) is not the depth-one gradient-descent limit, because its fired coefficients can change sign. Appendix A.8 proves the depth-one link at finite α.
- *Two layers and deeper.* In Alternating Gradient Flows [18], dormant neurons accumulate utility and activate at a threshold proportional to $\log(1/\alpha)$, without resets (Literature, from memory). Deeper attach moves are [Heuristic].

A.4 What training time controls

Proposition A.5 (Depth grows at most linearly in time (Proposition A2.1 in the derivation))**.** *[Proved] In either variant, a gate of depth D (the number of gates on the longest path from the inputs) fires only at times t with $\tau(t) \geq D\Theta$. Hence the depth of the circuit at time t is at most $\lfloor \tau(t)/\Theta \rfloor \leq \lfloor t/\Theta \rfloor$. In (LS), only finitely many events happen up to any time t.*

Proof sketch. A gate m fires at t_m with $\Theta = |S_m(t_m)| \leq \int_{a_m}^{t_m} \|r\|_n = \tau(t_m) - \tau(a_m)$. A gate of depth $D \geq 2$ has a source of depth $D-1$ that fired no later than a_m. Induct on D. □

Proposition A.6 (Training time does not bound the gate count uniformly over libraries (Proposition A2.2 in the derivation))**.** *[Proved] Consider the population race under uniform inputs, with $y = \mathrm{MAJ}(x_1, \ldots, x_m)$ for odd $m \leq d$, and let $u_m = \binom{m-1}{(m-1)/2} 2^{-(m-1)} = \sqrt{2/(\pi m)}\,(1 + o(1))$.*

(1) If $G = \{\mathrm{id}\}$, the m identity gates fire together at $t = \Theta/u_m$, and nothing fires earlier. So T jumps from 0 to $m \approx \frac{2}{\pi}(t/\Theta)^2$.

(2) For the toy library, exactly m^2 distinct functions (up to sign) fire together at $t = \Theta/u_m$: the m identities, and $\mathrm{AND}(x_i, x_j)$ and $\mathrm{AND}(\neg x_i, \neg x_j)$ for $i < j \leq m$. So T jumps to $m^2 \approx (\frac{2}{\pi})^2 (t/\Theta)^4$.

(3) Candidates that compute the same function up to sign and have the same availability time fire together. So the syntactic gate count is not bounded by any function of t/Θ uniformly over libraries. For a fixed finite library it is at most the number of syntactic candidates of depth at most $\lfloor \tau(t)/\Theta \rfloor$.

In particular, for the toy library no bound $T \leq C(t/\Theta)^2$ with C independent of d holds uniformly over the targets MAJ_m, $m \leq d$.

Proof sketch. MAJ_m is odd, so $\mathbb{E}y = \mathbb{E}[y x_i x_j] = 0$, and $\mathbb{E}[y x_i] = u_m$ for $i \leq m$ is the probability that the other votes tie. Expanding $\mathrm{AND}(\epsilon_1 a, \epsilon_2 b) = (\epsilon_1 a + \epsilon_2 b + \epsilon_1 \epsilon_2 ab - 1)/2$ shows that the listed gates have utility $\pm u_m$ and every other depth-one gate has utility at most $u_m/2$. The residual equals y until the first event, so the listed clocks reach Θ together while every other clock is at most $\Theta/2$. For (3), equal functions with equal availability times have identical clocks; adding J copies of the identity gate to G makes Jm syntactic gates fire in (1). □

On a sample, the race reproduces (2): for maj3 at $n = 4096$, 9 gates fire between $t/\Theta = 1.98$ and 2.12, against the population value $1/u_3 = 2$ [Numerical].

Theorem A.7 (Gate count times incoherence (Theorem A2.3 in the derivation)). *[Proved] For a set B of gates, let K_B be the centred empirical Gram matrix, $(K_B)_{mm'} = \langle g_m - \bar{g}_m, g_{m'} - \bar{g}_{m'} \rangle_n$, and let $\kappa_B = \lambda_{\max}(K_B)$.*

(1) (LS), any depth: at every time t, every set B of gates fired by time t satisfies $|B| \le \kappa_B \, (\tau(t)/\Theta)^2$. If every gate in B has availability time 0, $\tau(t)$ may be replaced by $\|R(t)\|_n \le \tau(t)$.

(2) (SC): at every time t, $|\mathrm{Sat}(t)| \le \kappa_{\mathrm{Sat}(t)} (\|R(t)\|_n/\Theta)^2$, and the same holds for $\operatorname{supp} c(t) \subseteq \mathrm{Sat}(t)$.

In particular $T(t) \le \kappa(t)(\tau(t)/\Theta)^2$, where $T(t)$ counts the distinct functions (up to sign) among the fired gates and $\kappa(t) = \kappa_B$ for one representative B of each. By Gershgorin's theorem $\kappa_B \le 1 + (|B| - 1)\mu_B$, where μ_B is the largest absolute centred covariance between two gates of B. So whenever $\mu_B(\tau/\Theta)^2 < 1$, $|B| \le (1 - \mu_B)(\tau/\Theta)^2/(1 - \mu_B(\tau/\Theta)^2)$.

Proof sketch. Each $m \in B$ has $\sigma_m S_m(t) = \Theta$. Because the intercept is refit, r has empirical mean zero, so $U_m = \langle r, \tilde{g}_m \rangle_n$ with $\tilde{g}_m = g_m - \bar{g}_m$. Let $B(s) = \{m \in B : a_m \le s\}$. Summing the clocks,

$$|B|\,\Theta = \int_0^t \Big\langle r(s), \sum_{m \in B(s)} \sigma_m \tilde{g}_m \Big\rangle_n ds \le \int_0^t \|r(s)\|_n \sqrt{\sigma^\top K_{B(s)} \sigma}\, ds \le \tau(t) \sqrt{\kappa_B |B|},$$

because a principal submatrix has no larger top eigenvalue. If all $a_m = 0$, then $S_m(t) = \langle R(t), \tilde{g}_m \rangle_n$, and Cauchy–Schwarz is applied once, with $\|R(t)\|_n$. The (SC) case is the same with $B = \mathrm{Sat}(t)$. □

The population analogue is tight up to a factor $\pi/2$: in Proposition A.6(1), $\kappa = 1$, $T = m$ and $(\tau/\Theta)^2 = 1/u_m^2 \approx \pi m/2$. A library-wide coherence does not help for the toy library. There $\mu = 1/2$ (for x_1 against $\mathrm{AND}(x_1, x_2)$), and the Gershgorin condition fails once $\tau \ge \sqrt{2}\,\Theta$.

Theorem A.8 (The readout's ℓ_1 norm grows at most linearly in time (Theorem A2.4 in the derivation)). *[Proved] In the (SC) variant on a depth-one dictionary, for every t,*

$$\Theta \sum_m |c_m(t)| = \langle R(t), f_t \rangle_n \le \|R(t)\|_n \sqrt{1 - \bar{y}^2} \le \tau(t), \qquad |c_0(t)| \le 1 + \|c(t)\|_1.$$

Proof sketch. All $a_m = 0$, so $S_m = \langle R, g_m \rangle_n$, and the KKT conditions give $c_m S_m = |c_m|\Theta$ for every m. Since $\langle R, 1 \rangle_n = 0$, summing gives $\Theta \|c\|_1 = \langle R, f_t - \bar{y} \rangle_n$. Now $f_t - \bar{y}$ is the projection of $y - \bar{y}$ onto a closed convex cone, so its norm is at most $\|y - \bar{y}\|_n = \sqrt{1 - \bar{y}^2}$. Apply Cauchy–Schwarz. This is LASSO duality, with $R(t)$ in the role of the residual and Θ in the role of the penalty. □

Checks [Numerical]. The toy races used $d = 30$, the toy library, n from 64 to 4096, the targets par2, maj3, and3, dnf3 and par3, and a test set of 20000 points (`race_bounds.txt` in `theory/checks`).

- Theorem A.2, Proposition A.5 and Theorem A.7 held at every event.
- Over all (LS) events, $\kappa \in [0.44, 4.59]$. The looseness $\kappa\tau^2/(\Theta^2 T)$ of Theorem A.7 is 1.00 on par2, where the bound is attained, and at most 4.96, 2.71 and 6.97 on maj3, and3 and dnf3. It reaches 400 on par3, where a single noise gate fires late.
- The identity of Theorem A.8 holds to $3 \cdot 10^{-14}$, including on 40 random correlated dictionaries with 130 exits from the saturated set, where the (SC) size bound holds with ratio at most 0.715.

- The Cauchy–Schwarz step of Theorem A.8 can be loose: maj3 at $n = 4096$ ends with $\|c\|_1 = 1.53$ against $\|R\|_n/\Theta = 15.2$.

A.5 Test error as a function of training time

Corollary A.9 (Test error along training (Corollary A2.5 in the derivation))**.** *[Proved for (a) and (a′); proved given the lemmas of Theorem A.3 for (b)] Let $h_t = \operatorname{sgn} f_t$ and $D(t) = \lfloor \tau(t)/\Theta \rfloor$. Each of (a), (a′) and (b) holds with probability at least $1 - \delta$, simultaneously for all $t \geq 0$.*

(a) Size $\leq$ incoherence $\times$ time^2 *((LS), any depth). Theorem A.2 (all items) holds for h_t with $T = T(t)$, and $T(t) \leq \lfloor \kappa(t)\tau(t)^2/\Theta^2 \rfloor$. Since L_T increases with T, $L_{T(t)}$ may be replaced by $L_{\lfloor \kappa\tau^2/\Theta^2 \rfloor}$. For example, at zero training error, $\operatorname{err}(h_t) \leq L_{\lfloor \kappa\tau^2/\Theta^2 \rfloor}/n$.*

(a′) Time-only count for a fixed library *((LS) at any depth, or (SC)). Let $\mathcal{F}_{\leq D}$ be the set of non-constant functions computed by gates of depth at most D over the library, counted as distinct up to sign on the sample. Let $\bar{\kappa}_D$ be the top eigenvalue of their centred empirical Gram matrix. Both depend only on the library and the unlabelled inputs. Suppose the race identifies candidates that coincide up to sign on the sample, and refits with one representative per class. Then Theorem A.2 (all items) holds for h_t with $T = T(t)$, and*

$$T(t) \leq \bar{T}(t) := \min\Big\{ |\mathcal{F}_{\leq D(t)}|,\ \Big\lfloor \bar{\kappa}_{D(t)} \frac{\tau(t)^2}{\Theta^2} \Big\rfloor \Big\}, \qquad \text{so } L_{T(t)} \leq L_{\bar{T}(t)}.$$

For a depth-one race, and for (LS) at any depth while $\tau(t) < 2\Theta$, $\tau(t)$ may be replaced by $\|R(t)\|_n$.

(b) Norm version, time only *((SC), depth one, $|\mathcal{F}_1| \leq |G|d^k$). The two bounds of Theorem A.3 hold with $B_j \leq 2\max(1, \tau(t)/\Theta)$. The gap is therefore $O((1 + \tau(t)/\Theta)\sqrt{\log(|G|d^k)/n})$, linear in training time and independent of the number of gates.*

Proof sketch. (a) Theorem A.2 holds on one event uniformly in T, and $h_t \in \mathcal{H}_{T(t)}$; then apply Theorem A.7. (a′) By Proposition A.5, every fired gate has depth at most $D(t)$. By the merging convention, the representatives form a subset of $\mathcal{F}_{\leq D(t)}$. Sign flips conjugate the Gram matrix by a diagonal ± 1 matrix, so K_B is a principal submatrix of the library Gram matrix, and $\kappa_B \leq \bar{\kappa}_{D(t)}$. (b) Combine Theorem A.3 with Theorem A.8. □

What the three versions give.

- *(a) is not a bound in terms of training time alone.* $\kappa(t)$ is a property of the learned circuit, and it can grow with T: in the population, $\kappa = 3$ for maj3 after its first event and 5.3 after its second. Because $T(t) \leq \lfloor \kappa\tau^2/\Theta^2 \rfloor$, (a) is never tighter than Theorem A.2 at the realized $T(t)$. Its content is explanatory.
- *(a′) is a genuine time-only bound, but weak.* For the toy library the population value of $\bar{\kappa}_1$ is exactly d, by a Parseval computation. The empirical values at $d = 30$ are $\bar{\kappa}_1 = 88, 58, 41, 35, 32$ at $n = 64, \ldots, 16384$. At zero training error and $\|R\|_n/\Theta = 1, 1.5, 1.99$, (a′) gives 0.21 ($\bar{T} = 34$), 0.41 and 0.61 at $n = 4096$, and 0.053 ($\bar{T} = 32$), 0.115 and 0.195 at $n = 16384$. For comparison, Theorem A.2 at the realized size gives 0.004 and 0.001 for par2, which fires a single gate at $\tau = \Theta$. So (a′) answers the original question literally (short training, so a small circuit, so a Chernoff bound). But it does so only at depth one, only for $\tau \lesssim 2\Theta$, and at the cost of a factor $\bar{\kappa}_1 \approx d$ in size. Without the merging convention it can fail: in the referee's example ($d = 6$, $n = 4$, par2), 24 gates that are distinct on $\mathcal{X}$ fire, against $\bar{T} = 2$.

- *(b) is near-vacuous at these sizes.* With $|\mathcal{F}| = 2205$ at $n = 4096$, Theorem A.3 bounds the clipped square loss by 0.43–0.92 over all events (the true values are 0–0.38), and the 0-1 error (ramp loss with $\theta = 1/2$) by 0.40–0.97. At $n = 256$ both bounds exceed 1. The norm route would need $n \gtrsim 10^5$ here.

Summing up, in the race model training time bounds the *depth*, the *incoherence-weighted size* and the depth-one *readout* ℓ_1 *norm*. It bounds the raw gate count only through an incoherence factor, either the learned circuit's $\kappa(t)$ or the library's $\bar{\kappa} \approx d$. No bound with a library-independent constant holds (Proposition A.6).

Proposition A.10 (No fitting of noise before $t \approx \Theta\sqrt{n}$ on parity-like targets (Proposition A2.6 in the derivation))**.** *[Proved] Suppose* $\mathbb{E}y = 0$ *and* $\mathbb{E}[yg] = 0$ *for every depth-one candidate* g*, as for* $y = \chi_A = \prod_{i\in A} x_i$ *with* $|A| > k$ *under uniform inputs. Let* M_1 *be the number of depth-one candidates and* $\epsilon = \sqrt{2\log(2(M_1+1)/\delta)/n}$*. With probability at least* $1-\delta$*, no gate fires before time* $\Theta/(2\epsilon)$*. Until then the predictor is the constant* $\operatorname{sgn}\bar{y}$*, with test error* $1/2$*.*

Here Hoeffding's inequality bounds every initial utility by 2ϵ. On par3, the first noise gate fires at $t/\Theta = 2.65, 5.16, 9.97, 20.0$ for $n = 64, 256, 1024, 4096$, against the proved lower bounds $0.84, 1.68, 3.35, 6.71$ [Numerical].

A.6 SGD and minibatches: reproducing the population run

Stability results are proved for a *greedy argmax model*, which adds one gate per step. In the *population greedy run* (PG), $C_0 = \emptyset$. Given C_t, let f_t be the $L^2(P)$ projection of y onto $V_t = \operatorname{span}\{1, g : g \in C_t\}$ and $r_t = y - f_t$. Each candidate m has utility $U_t(m) = \mathbb{E}[r_t g_m]$; candidates computing the same function up to sign are identified, and there are $M_t \le |G|(d+t)^k$ of them. PG adds $m_t = \arg\max_m |U_t(m)|$. The assumptions are:

- a gap $\gamma > 0$: $|U_t(m_t)| \ge |U_t(m)| + \gamma$ for all $m \ne \pm m_t$ and all $t < T$;
- $|r_t| \le R$ almost surely;
- $\lambda_* := \min_{t\le T} \lambda_{\min}(K_t) > 0$, where K_t is the $L^2(P)$ Gram matrix of $(1, C_t)$.

The *empirical greedy run* (EG) uses a batch B_t of b samples at step t. Either $B_t = S$ for all t (full batch), or the B_t are fresh independent batches. At each step EG refits by least squares on B_t and adds a candidate that maximizes $|\langle \hat{r}_t, g_m\rangle_{B_t}|$. Its final predictor is the sign of the least-squares refit over C_T on B_T.

Theorem A.11 (Decision stability (Theorem A3.1 in the derivation))**.** *[Proved] Let* $\epsilon := \gamma/(2(1+\sqrt{2T/\lambda_*}))$*, and suppose*

$$b \ge \frac{2R^2}{\epsilon^2}\log\frac{4\sum_{t<T}(M_t+t+1)}{\delta} \qquad \text{and} \qquad b \ge \frac{8(T+1)^2}{\lambda_*^2}\log\frac{4(T+1)^3}{\delta}.$$

Then, for both the full-batch and the fresh-batch protocol, with probability at least $1-\delta$*:*

(1) EG's first T *gates compute the same functions as* $m_0, \dots, m_{T-1}$*, so its circuit is* C_T*;*

(2) if $y \in V_T$ *almost surely (an exact representation), EG's final predictor equals* y *almost surely, so its test error is* 0*.*

Moreover, with probability at least $1-2\delta$*:*

(3) EG's final predictor lies in the data-independent class $\mathcal{H}(C_T)$ *of sign readouts over* C_T*, and the four bounds of Theorem A.2 hold for it with* $L = \rho_T + \log(1/\delta)$ *in place of* L_T*. These bounds are on the sample used for the final refit:* S *for full batch,* B_T *for fresh batches.*

Proof sketch. The functions r_t, g_m and K_t belong to the population run and do not depend on the data. So Hoeffding's inequality and a union bound give an event on which every empirical utility $\langle r_t, g\rangle_{B_t}$ is within ϵ of its mean, and every empirical Gram entry is within $\lambda_*/(2(T+1))$ of K_t; hence $\lambda_{\min}(\hat{K}_t) \geq \lambda_*/2$. By induction, suppose EG has C_t. Then $\hat{r}_t = (I - \hat{\Pi}_t)r_t$, and $\|\hat{\Pi}_t r_t\|^2_{B_t} < 2(t+1)\epsilon^2/\lambda_*$, because r_t is population-orthogonal to V_t. So every empirical utility is within $\gamma/2$ of $U_t(m)$, and the gap forces the empirical argmax to compute $\pm m_t$. Item (3) applies the single-class form of Theorem A.2 to $\mathcal{H}(C_T)$. □

Theorem A.12 (Decision stability with reuse (Theorem A3.5 in the derivation))**.** *[Proved] Assume the setting of Theorem A.11. Draw S of size n iid from P. At each step, let B_t consist of b indices drawn iid uniformly from $[n]$, independently of S and of the other steps; this is minibatch sampling with replacement, reusing S. Let $N_1 = \sum_{t<T}(M_t + t + 1)$, $N_2 = (T+1)^3$, and suppose*

$$(n^{-1/2} + b^{-1/2})R\sqrt{2\log(8N_1/\delta)} \leq \epsilon \qquad \text{and} \qquad (n^{-1/2} + b^{-1/2})\sqrt{2\log(8N_2/\delta)} \leq \frac{\lambda_*}{2(T+1)}.$$

Then, with probability at least $1-\delta$, items (1) and (2) of Theorem A.11 hold for the reuse run. With probability at least $1-2\delta$, the bounds of Theorem A.2 hold for its final predictor on S, with $L = \rho_T + \log(1/\delta)$.

The proof splits each deviation into a sample term over S and a conditional batch term, both union-bounded over the same data-independent functions. These conditions are those of Theorem A.11 with b replaced by $n_{\text{eff}} = (n^{-1/2} + b^{-1/2})^{-2} \geq \min(n, b)/4$ and δ by $\delta/2$.

The thresholds are far above the experiments [Numerical].

- *par2.* For par2 with XOR in G, $\gamma = 1/2$, $R = \lambda_* = 1$ and $M_0 = 2205$, so $\epsilon = 0.10355$, and Theorem A.11 needs $b \geq 2254$. Theorem A.12 needs $n = b \geq 9530$, or $b \geq 2383$ as $n \to \infty$, or $b \geq 6225$ at $n = 16384$. The experiments' $b = 64$ is covered at no n, while the toy race recovers par2 exactly at $n = 64$. In a Monte Carlo of the greedy first step, the right gate is the *unique* argmax in about 81–94% of runs at $b = 16$ and in all runs at $b = 64$.
- *The Gram condition can dominate.* With the toy library, $\delta = 0.05$, $\gamma = 1/2$, $R = 1$, $T = 30$ and $\lambda_* = 1/2$, the utility condition needs $7.9 \cdot 10^4$ samples and the Gram condition $4.5 \cdot 10^5$.
- *Comparison with Occam.* Stability gives exact recovery from about $\max\{TR^2/(\lambda_*\gamma^2), T^2/\lambda_*^2\}\log(MT/\delta)$ samples, with no $1/\varepsilon$, whereas the Occam bound needs about $T \log M/\varepsilon$ samples for error ε. But stability says nothing under ties ($\gamma = 0$), such as the 9-gate cluster of maj3.

For the *race* under SGD, the derivation proves only that clocks concentrate. Run the race in steps of size η with fresh batches of b iid samples, a predictable residual bounded by R, and predictable availability. Then every clock stays within $R\sqrt{2\eta t_0 \log(2|\mathcal{M}|/\delta)/b}$ of its drift clock up to time t_0, with probability at least $1-\delta$, over a fixed finite candidate set $\mathcal{M}$ (Lemma A3.2, [Proved]). Numerically, the observed maximum clock error is 75–88% of this bound. The claim that the SGD race fires the same gates as the population race also needs readout tracking and a crossing-time margin, and is [Heuristic]. For one-pass training, the derivation also proves an online-to-batch statement in the style of [9]: the minibatch loss of one-pass SGD, computed before each update, is a valid estimate of the trajectory's average test loss (Proposition A3.3, [Proved]).

A.7 Greedy growth versus minimum circuits

The *minimum-consistent-circuit* (MCC) learner outputs a smallest circuit, with the same readout class, that is consistent with S. Theorem A.2 gives err $\leq L_{s^*}/n$ for MCC in the realizable case,

where s^* is the size of a smallest consistent circuit, and err $\le L_{T_{\text{greedy}}}/n$ for the race when it is stopped. The comparison is entirely about T_{greedy} versus s^*.

Proposition A.13 (Equality on library targets (Proposition A4.1 in the derivation)). *[Proved] Let $y = \pm g^*$ for a depth-one candidate g^*, with $0 < P(y = 1) < 1$; no balance condition is needed. In the population race, in either variant, g^* and its duplicates fire first, at $t = \Theta/\mathrm{Var}(y)$, strictly before every other candidate. The refit then has zero residual and the race stops, with $T = 1 = s^*$. The same holds on a sample that contains both labels, with $\mathrm{Var}_n(y)$ in place of $\mathrm{Var}(y)$ and with "duplicates" meaning candidates equal to $\pm y$ on the sample.*

Proof sketch. For $|g| \le 1$, $|\mathbb{E}[(y-\bar{y})g]| \le \mathbb{E}|y-\bar{y}| = 1-\bar{y}^2 = \mathrm{Var}(y)$, with equality only for $g = \pm y$, since $y - \bar{y} \neq 0$ everywhere. So g^* has the largest possible utility, and every other candidate has a strictly smaller one. □

Proposition A.14 (Overshoot from ties and from square-loss refits (Proposition A4.2 in the derivation)). *[Proved (population)] Use the toy library under uniform inputs.*

(1) Ties. *For $y = \mathrm{maj}_3$, 9 distinct gates fire together at $t = 2\Theta$. After this event the refit is $(x_1 + x_2 + x_3)/2$, so the 0-1 error is already zero. Three identity gates suffice, and an exhaustive search finds no consistent circuit with one or two gates, so $s^* = 3$ [Numerical].*

(2) Square-loss refits keep going. *The residual is then $-x_1x_2x_3/2$, and no depth-one candidate ever fires again. Exactly 12 depth-two candidates (distinct up to sign) have the largest utility magnitude $1/4$, and all others have at most $1/8$. The 12 fire together at exactly 6Θ, after which y is represented exactly. So the population race ends at $t = 6\Theta$ with $9 + 12 = 21$ gates.*

(3) Overshoot by a factor m. *For $y = \mathrm{MAJ}_m$ (m odd), the first event fires m^2 distinct gates at $t = \Theta/u_m$, and the refit $u_m \sum_{i \le m} x_i$ already has zero 0-1 error. So $T_{\text{greedy}} = m^2$ while $s^* \le m$, and Theorem A.2 pays $L_{m^2} \approx m^2 \log(6d^2) + m^4 \log 2$ in place of L_m.*

Proof sketch. The 9 gates and 1 span exactly the functions of degree at most 2 in x_1, x_2, x_3, so the refit is the degree-$\le$ 2 part of $\mathrm{maj}_3 = \frac{1}{2}(x_1 + x_2 + x_3 - x_1x_2x_3)$. A depth-two candidate $\varphi(a, b)$ has utility $-\frac{1}{2}c_3\widehat{ab}(\{1,2,3\})$, where c_3 is the coefficient of ab in φ. A Fourier lemma gives $|\widehat{ab}(\{1,2,3\})| \le 1/2$, with equality for exactly 12 source pairs. Item (3) holds because MAJ_m has no degree-2 Fourier mass. □

Proposition A.15 (Parity is never found (Proposition A4.3 in the derivation)). *[Proved] Let $y = \chi_A$ with $|A| = s > k$, under uniform inputs.*

(1) In the population race no candidate ever fires, so $T(t) = 0$ and the test error is $1/2$ for all t.

(2) If $\mathrm{XOR} \in G$, a chain of $s-1$ XOR gates with the output-gate readout computes the target. So the MCC learner over that readout class, or any learner that outputs a consistent circuit of at most $s-1$ gates with that readout, has error at most L_{s-1}/n, with $L_{s-1} = W_{s-1} + \log 2 + \log \frac{s(s+1)}{\delta} = O(s \log(d+s) + \log(1/\delta))$.

(3) On a sample, Proposition A.10 applies.

Item (1) holds because every depth-one candidate depends on at most $k < s$ variables, so it is uncorrelated with χ_A. This is the circuit-leap obstruction in its simplest form.

Overshoot in the toy runs [Numerical].

- *maj3.* At $n \geq 256$ the race reaches zero training error with 6 gates, against $s^* = 3$, and ends with 10. On a sample, noise breaks the 12-way tie, and the first of the 12 completes the representation.
- *dnf3.* The race needs 18 gates, while a 6-gate circuit exists. At $n = 4096$, Theorem A.2 gives 0.084 with $T = 18$, where $T = 6$ would give 0.019.
- *par3.* Only noise gates fire, and the test error is 0.50. Theorem A.2 stays valid there but gives 0.52–0.95, which certifies nothing.

Each gate of the greedy race must beat the noise level $\sqrt{\log M/n}$ at the moment it is added, whereas the MCC learner pays for its size only once [Heuristic].

A.8 Diagonal linear networks: a rigorous instance (Section C2)

Setting. The dictionary is $\{\chi_J : J \in \mathcal{J}\}$ with $|\mathcal{J}| = D$, for example all $J \subseteq [d]$ with $|J| \leq k$, so that $D = D_k := \sum_{i \leq k} \binom{d}{i}$. Let $X \in \{\pm 1\}^{n \times D}$ with $X_{iJ} = \chi_J(x_i)$, and $K = X^\top X/n$. The model is $f_w = \sum_J w_J \chi_J$, with loss $L(w) = \frac{1}{2n}\|Xw - y\|^2$ and parametrization $w = u \odot u - v \odot v$. It is trained by gradient flow on (u, v) from $u(0) = v(0) = \alpha \mathbf{1}$. Write $r = y - Xw$ and $g = \nabla L(w) = -X^\top r/n$, so that $g_J = -\langle r, \chi_J \rangle_n$, and let $G(t) = \int_0^t g$, $\tau(t) = \int_0^t \|r\|_n$ and $R(t) = \int_0^t r$. Define

$$\Theta_\epsilon := \tfrac{1}{4} \operatorname{arsinh} \frac{\epsilon}{2\alpha^2} \;\geq\; \tfrac{1}{4} \log \frac{\epsilon}{\alpha^2}.$$

Lemma A.16 (Closed form (Lemma C2.1 in the derivation))**.** *[Proved] The flow exists and is unique for all $t \geq 0$, and $u(t) = \alpha e^{-2G(t)}$, $v(t) = \alpha e^{2G(t)}$ and $w(t) = -2\alpha^2 \sinh(4G(t))$. Equivalently, w follows the mirror flow of the hyperbolic-entropy potential. Moreover $L(w(t))$ is nonincreasing, $\|r(t)\|_n \leq \|y\|_n$, and $|g_J(t)| \leq \|r(t)\|_n$.*

The hyperbolic entropy and its implicit bias are due to Woodworth et al. (COLT 2020). The lemma follows because, for given g, $\dot{u} = -2g \odot u$ and $\dot{v} = 2g \odot v$ are linear. So $-G_J(t) = \int_0^t \langle r, \chi_J \rangle_n$ is exactly the race clock, and $|w_J| \geq \epsilon$ is exactly the event that the clock has reached $\Theta_\epsilon \approx \frac{1}{2}\log(1/\alpha)$. The results below are Theorems A.7 and A.8 for the actual gradient flow at finite α, with "fired" replaced by "grown above ϵ".

Theorem A.17 (No early growth; first escape (Theorem C2.2 in the derivation))**.** *[Proved]*

(1) For all t and J, $|G_J(t)| \leq \tau(t) \leq t\|y\|_n$, so $|w_J(t)| \leq 2\alpha^2 \sinh(4\tau(t))$. In particular, no coordinate reaches ϵ while $\tau(t) < \Theta_\epsilon$.

(2) Let $U_0 = \max_J |\langle y, \chi_J \rangle_n|$, $\epsilon_D = \sqrt{\lambda_{\max}(K) D}\, \epsilon \leq D\epsilon$, and $t_\epsilon = \inf\{t : \max_J |w_J(t)| \geq \epsilon\}$. If $U_0 > \epsilon_D$, then $\Theta_\epsilon/(U_0 + \epsilon_D) \leq t_\epsilon \leq \Theta_\epsilon/(U_0 - \epsilon_D)$.

Before t_ϵ, $\|Xw\|_n \leq \epsilon_D$, so every utility stays within ϵ_D of its initial value. Item (2) is the rate law "escape time $\approx \log(1/\alpha)/U$" with explicit two-sided constants.

Theorem A.18 (How many coordinates can have grown (Theorem C2.3 in the derivation))**.** *[Proved] Let $B = B_\epsilon(t) = \{J : |w_J(t)| \geq \epsilon\}$, $N_\epsilon(t) = |B|$, and $\kappa_B = \lambda_{\max}(X_B^\top X_B/n)$. Then:*

(1) $N_\epsilon(t) \leq \big(\int_0^t \|\nabla L(w(s))\|_2\, ds\big)^2/\Theta_\epsilon^2$;

(2) $N_\epsilon(t) \leq \kappa_B\, \tau(t)^2/\Theta_\epsilon^2$. With the empirical coherence $\mu_n = \max_{J \neq J'} |\langle \chi_J, \chi_{J'} \rangle_n|$ and $x = (\tau/\Theta_\epsilon)^2$, whenever $\mu_n x < 1$ this gives $N_\epsilon \leq (1 - \mu_n)x/(1 - \mu_n x)$. For uniform inputs and all characters of degree at most k, $\mu_n \leq \bar{\mu} := \sqrt{2\log(2D_{\min(2k,d)}/\delta')/n}$ with probability at least $1 - \delta'$;

(3) $\sum_J |w_J(t)||G_J(t)| = \langle R(t), f_{w(t)}\rangle_n$, *and hence* $\sum_{J\in B} |w_J(t)| \le \tau(t)\|f_{w(t)}\|_n/\Theta_\epsilon \le 2\tau(t)/\Theta_\epsilon$.

Proof sketch. Every $J \in B$ has $|G_J(t)| \ge \Theta_\epsilon$. Item (1) follows from Minkowski's integral inequality. For item (2), with $\sigma_J = -\operatorname{sgn} G_J$, $N\Theta_\epsilon \le \sum_{J\in B}|G_J| = \int_0^t \langle r, \sum_{J\in B}\sigma_J\chi_J\rangle_n \le \tau(t)\sqrt{\sigma^\top K_B \sigma} \le \tau(t)\sqrt{\kappa_B N}$, exactly as in Theorem A.7. The coherence bound is Hoeffding's inequality over the characters $\chi_{J\triangle J'}$. Item (3) follows from $\operatorname{sgn} w_J = -\operatorname{sgn} G_J$. □

Item (2) is exact at the first crossing when $y = \chi_{J^*}$ is empirically orthogonal to the rest of the dictionary (for example, on the full cube). At nonzero coherence it becomes tight as $\alpha \to 0$.

Theorem A.19 (Generalization, uniform in time and threshold (Theorem C2.4 in the derivation))**.** *[Proved] Let* $\ell_N := \log\binom{D}{N} + \log\Phi(2^{\min(N,d+1)}, N)$, *and let* $\mu_\epsilon(t) := \sum_{J\notin B_\epsilon(t)}|w_J(t)| \le (D - N_\epsilon(t))\epsilon$. *With probability at least* $1-\delta$ *over* S, *simultaneously for all* $t \ge 0$ *and all* $\epsilon > 0$, *writing* $N = N_\epsilon(t)$,

$$\operatorname{err}(\operatorname{sgn} f_{w(t)}) \le \operatorname{kl}_+^{-1}\Big(\widehat{P}_S[y f_{w(t)}(x) \le 2\mu_\epsilon(t)],\ \frac{\ell_N + \log\frac{(N+1)(N+2)}{\delta}}{n}\Big).$$

The bound holds for $P(y f_{w(t)}(x) \le 0)$, *which is at least the test error under either convention for* $\operatorname{sgn}(0)$.

Proof sketch. Split $f_{w(t)} = f_L + f_s$, with f_L supported on $B_\epsilon(t)$ and $\|f_s\|_\infty \le \mu_\epsilon(t)$. The set $\{y f_L \le \mu\}$ is an affine threshold of the vector $(y\chi_J(x))_{J\in B}$, which takes at most $2^{\min(N,d+1)}$ values. So by Cover's count these sets number at most e^{ℓ_N} for each N. Apply kl-Chernoff with a union over these sets and over N. Then use $yf \le 0 \Rightarrow y f_L \le \mu$ on test points, and $y f_L \le \mu \Rightarrow yf \le 2\mu$ on training points. □

Corollary A.20 (The time-indexed form (Corollary C2.5 in the derivation))**.** *[Proved] Fix* $\gamma \in (0, 1]$, *and put* $\epsilon = \gamma/(2D)$, $\Theta_\gamma := \Theta_\epsilon = \frac{1}{4}\operatorname{arsinh}\frac{\gamma}{4D\alpha^2}$, $x(t) = (\tau(t)/\Theta_\gamma)^2$ *and* $\ell_N^* = \max_{N'\le N}\ell_{N'}$. *Let* $\bar N(t) \le D$ *be any quantity, possibly data-dependent, with* $N_\epsilon(t) \le \bar N(t)$. *On the event of Theorem A.19, for all* t *and* γ,

$$\operatorname{err}(\operatorname{sgn} f_{w(t)}) \le \operatorname{kl}_+^{-1}\Big(\widehat{P}_S[y f_{w(t)} \le \gamma],\ \frac{\ell^*_{\bar N} + \log\frac{(\bar N+1)(\bar N+2)}{\delta}}{n}\Big).$$

There are three admissible choices of $\bar N$, *from most to least data-dependent:*

(1) $\bar N_\kappa = \min\{D, \lfloor \kappa_B x(t)\rfloor\}$, *with the measured* κ_B *of the grown set;*

(2) if $\mu_n x(t) < 1$, $\bar N_G = \min\{D, \lfloor (1-\mu_n)x/(1-\mu_n x)\rfloor\}$;

(3) for uniform inputs and all characters of degree at most k, *with* $\bar x$ *either* $x(t)$ *or* $(t/\Theta_\gamma)^2$: *if* $\bar\mu\bar x < 1$, $\bar N_H = \min\{D, \lfloor(1-\bar\mu)\bar x/(1-\bar\mu\bar x)\rfloor\}$, *with total probability at least* $1-\delta-\delta'$.

Only choice (3) makes the size term a function of time alone, through (n, D, τ) or (n, D, t). It needs $n > 2\log(2D_{\min(2k,d)}/\delta')\,(\tau/\Theta_\gamma)^4$: the sample size must grow like the fourth power of the effective time. The dictionary size enters only logarithmically, both in the capacity $\ell_N \le N\log\frac{eD}{N} + N\min(N, d+1)\log 2 + \log 2$ and in the unit of time $\Theta_\gamma \approx \frac{1}{2}\log\frac{1}{\alpha} - \frac{1}{4}\log\frac{4D}{\gamma}$.

Discrete GD and SGD (Proposition C2.6 in the derivation) [Proved]. Let g_j be any gradient estimates: full-batch, or minibatch, fresh or reused. Take steps $u_{j+1} = u_j \odot (1 - 2\eta g_j)$ and $v_{j+1} = v_j \odot (1 + 2\eta g_j)$, let $\bar G = \max_{j<K}\|g_j\|_\infty$, assume $2\eta\bar G < 1$, and let $c_\eta = 1/(1 - 4\eta^2\bar G^2)$.

- *No early growth.* No coordinate reaches ϵ while the minibatch effective time $\tau_K^{\mathrm{b}} = \eta\sum_{j<K}\|r_j\|_{B_j}$ is below $\Theta_\epsilon(1 - 4\eta^2\bar G^2)$.

- *Count under any batches.* $N_\epsilon(K) \le (c_\eta \eta \sum_{j<K} \|g_j\|_2)^2/\Theta_\epsilon^2$. This uses gradient norms, not losses, and can be larger than the loss-based quantity by up to a factor $\sqrt{D}$.
- *Count under full batch.* $N_\epsilon(K) \le \kappa_B \tau_K^2/(\Theta_\epsilon - E_K)^2$ whenever $\Theta_\epsilon > E_K$, where $\tau_K = \eta \sum_{j<K} \|r_j\|_n$ and $E_K = \frac{1}{2}\sum_{j<K} \frac{(2\eta\|g_j\|_\infty)^3}{3(1-(2\eta\|g_j\|_\infty)^2)}$.
- *Generalization.* Theorem A.19 holds verbatim for the iterates of any algorithm, and so does Corollary A.20 with these counts.

Checks [Numerical]. The instance has $d = 10$ and $k = 3$, so $D = 176$, with $n = 100$ or 400; test errors are exact over the cube.

- The closed form holds to about 10^{-8}, the integration tolerance. Theorem A.17 held in all 40 cases.
- In Theorem A.18(2) the worst ratio $N/(\kappa_B\tau^2/\Theta^2)$ is 0.33–0.94, and 1.00 for par2 at $\alpha = 10^{-8}$.
- Under GD and SGD the product form holds to 10^{-13}. The count ratios are 0.62–0.80 for the full-batch bound, but only 0.07–0.12 for the SGD bound.

Table 5 gives the bounds at the end of training ($t = 60$), at $\gamma = 1/2$, with $\bar{x} = x(t)$ in choice (3).

Table 5: Generalization bounds in the diagonal network ($d = 10$, $D = 176$, $t = 60$). "vac." means vacuous. Source: `c2_diag_check.txt` and `C_r2_checks.out` in `theory/checks`.

target	n	α	test err	$N_{10^{-2}}$	Theorem A.19	C2.5(1), measured κ_B	C2.5(3), Hoeffding $\bar{\mu}$
par2	100	10^{-3}	0	1	0.107	0.218	vac.
maj3	100	10^{-3}	0	4	0.275	0.810	vac.
maj3	400	10^{-3}	0	4	0.077	0.316	vac.
and3	400	10^{-3}	0	8	0.166	0.477	vac.
maj5	100	10^{-3}	0.073	48	0.951	0.998	vac.
par4	100	10^{-3}	0.516	73	0.988	1.0	vac.
par2	100	10^{-8}	0	1	0.107	0.107	0.163
and3	400	10^{-8}	0	8	0.166	0.304	vac.

- The a-posteriori bound (Theorem A.19) is non-vacuous for targets that are sparse in the dictionary.
- The time-indexed bound with the measured κ_B gives 0.11–0.48 for the sparse targets at $n = 400$ and for par2 at $n = 100$, but 0.81 for maj3 at $n = 100$.
- The forms without κ_B are *vacuous except for par2 at* $\alpha = 10^{-8}$. There they give 0.107 with the empirical μ_n and 0.163 with $\bar{\mu}$, and 0.275 with raw time t in place of τ, at $t = 10$ only.
- maj5 lies outside the dictionary, so coordinates keep growing. par4 is uncorrelated with every dictionary element, so only noise coordinates grow, and the bound correctly certifies nothing.

In this toy, then, the time-only form of the argument "short training, so few grown coordinates, so a Chernoff bound" works only for a single-character target at tiny α. The form with the measured κ_B works for every sparse target.

A.9 Conditional transfer to ReLU networks (Section C3)

What is known (Literature). For small initialization, rigorous results cover early alignment and the first escape in two-layer and homogeneous networks, for example [8]. Exact saddle-to-saddle

limits are proved only for diagonal networks [18, 22]; for general two-layer ReLU networks, AGF [18] is a model with matching experiments. Layer balancedness is due to [12]. The derivation found no non-asymptotic theorem that a ReLU network tracks a greedy process beyond the first escape, and none at μP initialization. What it proves for small-initialization gradient descent is one-sided: upper bounds on growth. The study's networks (two hidden layers with a hidden bias, μP scale, Adam with per-layer multipliers) are outside both lemmas below.

Lemma A.21 (Growth bounds (Lemmas C3.1 and C3.2 in the derivation)). *[Proved] Let $\tilde{x} = (x, 1)$ and $\lambda_X = \lambda_{\max}(\frac{1}{n}\sum_i \tilde{x}_i\tilde{x}_i^\top)$.*

(1) For a two-layer network $f(x) = \sum_j a_j \mathrm{ReLU}(w_j \cdot \tilde{x})$ under gradient flow (with any Clarke selection), $\rho_j = a_j^2 + \|w_j\|^2$ satisfies $\rho_j(t) \le \rho_j(0)e^{2\sqrt{\lambda_X}\tau(t)}$. Under gradient descent, $\rho_j^K \le \rho_j^0 e^{2\sqrt{\lambda_X}\tau_K}$ with $\tau_K = \eta\sum_{k<K}\|r_k\|_n$. Consequently the training loss cannot fall below $\frac{1}{2}(\|y\|_n - \epsilon)^2$ while $\tau(t) < \frac{1}{2\sqrt{\lambda_X}}\log\frac{2\epsilon}{\sqrt{\lambda_X}\sum_j \rho_j(0)}$.

(2) For a bias-free deep ReLU network $f = W_L\sigma(\cdots\sigma(W_1\tilde{x}))$ with $L \ge 3$, under gradient flow, all $\|W_l\|_F^2$ grow by the same amount $\Delta(t)$, and $Q(t) = \max_l \|W_l(0)\|_F^2 + \Delta(t)$ satisfies $Q(t)^{1-L/2} \ge Q(0)^{1-L/2} - (L-2)\sqrt{\lambda_X}\,\tau(t)$. With layers of Frobenius norm of order α, $\|f_t\|_n < \epsilon$, so the training loss stays at least $\frac{1}{2}(\|y\|_n - \epsilon)^2$, as long as $\tau(t) < (Q(0)^{-(L-2)/2} - (\epsilon/\sqrt{\lambda_X})^{-(L-2)/L})/((L-2)\sqrt{\lambda_X})$, which is of order $\alpha^{-(L-2)}$.

Both lemmas follow from the per-unit Euler identity $\frac{d}{dt}\|W_l\|_F^2 = 2\langle r, f\rangle_n$ and Gronwall comparison. They are the rigorous one-sided forms of "no unit grows before about $\log(1/\alpha)/U$" and "fresh chains cost $\alpha^{-(L-2)}$". Numerically, in (1) at $\alpha = 10^{-4}$, the escape happens at $t = 8.3$–8.7 for maj3 and 15.5–16.2 for par2, against the proved lower bound 3.7. In (2) at $L = 3$, the escape effective times are $65, 200, 673$ at $\alpha = 0.03, 0.01, 0.003$, against the bounds $3.0, 13.0, 47.8$. The scaling is right but the bound is loose by 14–21×. At $L = 4$ and $\alpha \ge 0.1$ the bound is vacuous [Numerical].

The hypothesis. Let $\theta(t)$ be the network's training trajectory on S, $\mathcal{T}$ a finite set of checkpoints, and P_X the input marginal of P. Let M be the (LS) race on the same S, with a finite library G (Boolean or real-valued) and threshold Θ. Write $h_s^M = \mathrm{sgn} f_s^M$ for its predictor at race time s, $T^M(s)$ for its gate count and $\tau^M(s)$ for its effective time, and let $s : \mathcal{T} \to [0, \infty)$ be a clock calibration. The *saddle-to-saddle approximation hypothesis* (SSA) at checkpoint t with tolerance $\eta(t)$ has two forms:

- the sign form, $\mathrm{dis}(\mathrm{sgn} f_{\theta(t)}, h_{s(t)}^M) := P_X(\mathrm{sgn} f_{\theta(t)} \ne h_{s(t)}^M) \le \eta(t)$;
- the L^2 form, $\|\mathrm{clip} f_{\theta(t)} - \mathrm{clip} f_{s(t)}^M\|_{L^2(P_X)} \le \eta(t)$, where clip clips to $[-1, 1]$.

For a Boolean library let $L_T^{\mathrm{sgn}} = W_T + \log\Phi(2^{\min(T,d)}, T)$ and $L_T^{\mathrm{mar}} = W_T + \log\Phi(2^{\min(T+1,d+1)}, T+1)$. For a real-valued library let $L_T^{\mathrm{sgn}} = W_T + \log\Phi(2^d, T)$ and $L_T^{\mathrm{mar}} = W_T + \log\Phi(2^{d+1}, T+1)$.

Theorem A.22 (The network inherits the race bound (Theorem C3.4 in the derivation)). *[Conditional: proved under hypothesis SSA at the checkpoints where it is invoked] There is an event of probability at least $1-\delta$ over S, not depending on the network, on which the following holds for every checkpoint $t \in \mathcal{T}$. Write $s = s(t)$ and $T = T^M(s)$; recall that $T \le \kappa^M(s)(\tau^M(s)/\Theta)^2$ by Theorem A.7.*

(1) If SSA holds at t in sign form, then $\mathrm{err}(\mathrm{sgn} f_{\theta(t)}) \le \eta(t) + \mathrm{kl}_+^{-1}(\widehat{\mathrm{err}}_S(h_s^M), (L_T^{\mathrm{sgn}} + \log\frac{2(T+1)(T+2)}{\delta})/n)$.

(2) If SSA holds at t in L^2 form, then for every $\gamma \in (0, 1)$, $\mathrm{err}(\mathrm{sgn} f_{\theta(t)}) \le \eta(t)^2/\gamma^2 + \mathrm{kl}_+^{-1}(\widehat{P}_S[y f_s^M \le \gamma], (L_T^{\mathrm{mar}} + \log\frac{2(T+1)(T+2)}{\delta})/n)$.

Proof sketch. Build, before training, one Occam event (as in Theorem A.2) for all threshold-readout circuit predictors and one for all margin sets $\{y(c_0 + \sum_m c_m g_m) \le \theta\}$, each at level $\delta/2$. Counting the image points of the gate vector gives the Boolean and the real-valued constants. In the sign form, $\mathrm{err}(\mathrm{sgn} f_\theta) \le \mathrm{err}(h_s^M) + \mathrm{dis}$. In the L^2 form, $y f_\theta \le 0$ implies $y \,\mathrm{clip} f^M \le |\mathrm{clip} f^M - \mathrm{clip} f_\theta|$, and Chebyshev's inequality bounds $P(|\mathrm{clip} f^M - \mathrm{clip} f_\theta| > \gamma) \le \eta^2/\gamma^2$. □

All the dynamics are in the hypothesis, and the theorem only converts it. Its gate count is time-indexed with a data-dependent κ^M, not a function of training time alone.

Corollary A.23 (Testing SSA, and the unconditional certificate (Corollary C3.5 in the derivation))**.** *[Proved] Let U be m unlabelled inputs, independent of the training data and the training randomness, and let the calibration be chosen from a finite grid Σ, possibly using U.*

(1) With probability at least $1-\delta_2$, for all $t \in \mathcal{T}$ and $s \in \Sigma$, $\mathrm{dis}(\mathrm{sgn} f_{\theta(t)}, h_s^M) \le \mathrm{kl}_+^{-1}(\widehat{\mathrm{dis}}_U, \log(2|\mathcal{T}||\Sigma|/\delta_2)/m)$. The squared clipped L^2 distance is at most its empirical value on U plus $4\sqrt{\log(2|\mathcal{T}||\Sigma|/\delta_2)/(2m)}$. So the sign form of SSA can be verified at every checkpoint.

(2) With no hypothesis at all, the distillation certificate (Theorem C1 in the derivation), applied to the candidates $\{h_s^M : s \in \Sigma\}$, bounds $\mathrm{err}(\mathrm{sgn} f_{\theta(t)})$ with probability at least $1 - \delta_1 - \delta_2$. The bound is a kl-Chernoff term for $\widehat{\mathrm{err}}_S(h_s^M)$ plus a kl-Chernoff term for $\widehat{\mathrm{dis}}_U(\mathrm{sgn} f_{\theta(t)}, h_s^M)$.

So SSA is not needed for the a-posteriori bound. It is needed only for the predictive, time-indexed statement.

A first test of SSA [Numerical]. The race uses the toy library, on the same training set as each saved μP Adam network. At 15 checkpoints, the network's signs on 4096 held-out inputs are compared with every race state.

- *First plateau.* For and3 and dnf3, the network predicts the majority label everywhere, which is the race's constant state. Any learner that fits the mean first does this, so it is no evidence for circuit growth.
- *Transitions.* For dnf3, mux and stair5, the best disagreement with a race state is 0.1–0.3, which is 40–80% of the test error. For maj3, majpar and parmaj, the network is no closer to any race state than to the target.
- *par3.* The fan-in-2 race never fires a useful gate, while the network learns par3 by step 500.

In this crude sign form, SSA therefore holds only in the trivial constant state, and Theorem A.22 cannot yet be applied to the study's runs. The valid, non-vacuous certificates for these networks come from Theorem C1 directly. Proving SSA for some ReLU class beyond the first escape is the main open problem of this part.

B Bounds for the trained networks

This appendix collects the generalization bounds proved for actual networks, as opposed to the discrete race model. They come from three derivations: `theory/B-nn-time-bounds.md` (parameter-space bounds for the study's networks), `theory/C-bridge-and-certificates.md` (distillation certificates and the online protocol) and `theory/D-early-phase-theorem.md` (early-phase theorems for two-layer networks from small initialization). Each derivation was refereed twice, and the second pass found only minor issues; the reports are `theory/checks/referee_B_r2.json`, `theory/checks/referee_C_r2.json` and `theory/checks/referee_D_r2.json`. Every result keeps

its number from the derivation, given in parentheses. The status labels mean the following. [Proved]: the derivation contains a complete proof with explicit constants, possibly invoking a named published result. [Conditional]: proved under a stated hypothesis that is not itself proved. [Conjecture]: not proved. [Numerical]: a proved bound evaluated by a script whose output is in `theory/checks/`. The proof sketches give only the key idea.

Scope. Appendices B.2 to B.6 concern the study's networks (three weight layers, μP initialization, SGD, GD or Adam). Appendices B.7 and B.8 concern two-layer networks trained from small initialization. None of D's theorems covers the study's μP Adam networks.

B.1 Notation

Data, errors, kl. Inputs are uniform on $\{\pm 1\}^d$ and labels are $y = f^\star(x) \in \{\pm 1\}$. The sample $S = ((x_i, y_i))_{i \le n}$ is iid. For predictors $h, g : \{\pm 1\}^d \to \{\pm 1\}$ put $\mathrm{err}(h) = \Pr(h(x) \neq y)$ and $\mathrm{dis}(h, g) = \Pr_x(h(x) \neq g(x))$. Hats denote empirical versions on S or on an unlabelled sample U, and $\widehat{P}_S[\cdot]$ is an empirical frequency on S. Write $\langle u, v\rangle_n = \frac{1}{n}\sum_i u(x_i)v(x_i)$ and $\|u\|_n^2 = \langle u, u\rangle_n$. For real f, $\mathrm{err}(\mathrm{sgn} f) \le \Pr(yf(x) \le 0)$, and the margin statements bound the right-hand side. $\mathrm{kl}(q\|p)$ is the binary relative entropy, and

$$\mathrm{kl}_+^{-1}(q, c) = \sup\{p \in [q, 1] : \mathrm{kl}(q\|p) \le c\} \;\le\; q + \sqrt{2qc} + 2c.$$

This inverse is nondecreasing in both arguments. Logarithms are natural.

The study's network. Here $d = 30$ and $f_\theta(x) = a \cdot \sigma(W_2\,\sigma(W_1 x + b_1) + b_2)$, with $\sigma = \mathrm{relu}$ and hidden widths $N = 128$.

- *Parameters.* The groups $g = 1, 2, 3$ are (W_1, b_1), (W_2, b_2) and a, with p_g entries ($p = 20608$ in total).
- *Initialization.* The μP initialization θ_0 has $W_1 \sim \mathcal{N}(0, 1/d)$, $W_2 \sim \mathcal{N}(0, 1/N)$, $a \sim \mathcal{N}(0, 1/N^2)$ and zero biases. It is drawn independently of S.
- *Losses.* $L_S(\theta) = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$, $L_0 = L_S(\theta_0) \approx 0.5$, and $R(\theta) = \Pr[yf_\theta(x) \le 0]$.
- *Optimizers.* Every update has the form $\theta_{k+1} = \theta_k - \eta M u_k$, where $M = \mathrm{diag}(m_g)$ holds the per-group μP multipliers. SGD ($\eta = 0.2$, batch 64) and full-batch GD use gradients, with $m = (2, 1, \frac{1}{2})$. Adam ($\eta = 10^{-3}$) uses $u_k = \hat{m}_k/(\sqrt{\hat{v}_k} + \epsilon)$, with $m = (1, \frac{1}{2}, \frac{1}{2})$.
- *Time and norms.* Training time is $t_K = \eta K$. The norms are $\|v\|_M^2 = v^\top M v$ and $\|v\|_{M^{-1}}^2 = v^\top M^{-1} v$.

B.2 Displacement from initialization as a function of time

Theorem B.1 (Gradient flow; Theorem B1.1 in the derivation)**.** *[Proved] Let $\theta : [0, t] \to \mathbb{R}^p$ be absolutely continuous with $\theta'(s) = -Mv(s)$ for a.e. s, where $v(s)$ is a Clarke subgradient of L_S at $\theta(s)$. Assume the chain rule $\frac{d}{ds} L_S(\theta(s)) = \langle v(s), \theta'(s)\rangle$ a.e. (It is automatic for C^1 losses. For ReLU networks with square loss the derivation takes it from Davis–Drusvyatskiy–Kakade–Lee, FoCM 2020, quoted from memory.) Put $E_g(t) = \int_0^t m_g \|v_g\|^2 ds$. Then:*

(1) $\sum_g E_g(t) = L_S(\theta(0)) - L_S(\theta(t))$;

(2) $\|\theta(t) - \theta(0)\|_{M^{-1}} \le \Lambda(t) := \int_0^t \|\theta'\|_{M^{-1}} \le \sqrt{t\,(L_S(\theta(0)) - L_S(\theta(t)))} \le \sqrt{tL_0}$;

(3) $\|\theta_g(t) - \theta_g(0)\|^2 \le m_g\, t\, E_g(t)$ *for every group g.*

Sketch. $\|\theta'\|_{M^{-1}}^2 = v^\top M v = -\frac{d}{ds} L_S$, which gives (1). Then apply the triangle inequality and Cauchy–Schwarz in s.

Corollary B.2 (Discrete time; Corollary B1.2). *[Proved] (a) For any vectors u_k, $\|\theta_K - \theta_0\|_{M^{-1}} \le \Lambda_K := \eta \sum_{k<K} \|u_k\|_M \le \sqrt{t_K E_K}$, where $E_K = \eta \sum_{k<K} \|u_k\|_M^2$. Per group, $\|\theta_{K,g} - \theta_{0,g}\|^2 \le m_g t_K E_{K,g}$ with $E_{K,g} = \eta m_g \sum_k \|u_{k,g}\|^2$. (b) For GD, suppose that for all $k < K$*

$$L_S(\theta_{k+1}) \le L_S(\theta_k) - c\,\eta\,\|\nabla L_S(\theta_k)\|_M^2, \qquad c > 0. \tag{D}$$

Then $E_K \le (L_0 - L_S(\theta_K))/c$, and so $\|\theta_K - \theta_0\|_{M^{-1}} \le \sqrt{t_K (L_0 - L_S(\theta_K))/c}$.

Under β-smoothness in the M-geometry, (D) holds with $c = 1 - \beta\eta/2$. For ReLU networks no global β exists, so (D) is an assumption about the trajectory. It can be checked a posteriori.

Proposition B.3 (SGD in expectation, smooth case; Proposition B1.3). *[Proved] Assume that L_S is β-smooth in the M-geometry on a convex set containing the iterates, and that $\eta \le 1/\beta$. Assume that each minibatch gradient $\hat{g}_k$ averages B per-example gradients at indices drawn uniformly with replacement, independently of the past, and that $\mathbb{E}[\|\hat{g}_k - \nabla L_S(\theta_k)\|_M^2 \mid \mathcal{F}_k] \le \nu^2/B$. Then, with $t = \eta K$,*

$$\mathbb{E}\|\theta_K - \theta_0\|_{M^{-1}}^2 \le 4t\,(L_0 - \mathbb{E}L_S(\theta_K)) + 2\eta t \nu^2 (1 + \beta t)/B.$$

Sketch. Split the displacement into drift and martingale noise. The cross terms of the noise vanish in expectation, and the drift is controlled by Cauchy–Schwarz and the descent lemma. The bound is weaker than Theorem B.1 by a factor of 4 and by a noise term that is quadratic in t. Its smoothness assumption fails for ReLU networks.

Lemma B.4 (Adam's exact worst-case step; Lemma B1.4). *[Proved] Let $0 \le \beta_1 < 1$ and $0 < \beta_2 < 1$, with $r := \beta_1^2/\beta_2 < 1$. For Adam's bias-corrected moments $\hat{m}_t, \hat{v}_t$ (coordinatewise, $m_0 = v_0 = 0$), every gradient sequence and every $t \ge 1$ satisfy*

$$\frac{|\hat{m}_t|}{\sqrt{\hat{v}_t}} \le C_t := \frac{1-\beta_1}{1-\beta_1^t}\sqrt{\frac{1-\beta_2^t}{1-\beta_2}}\sqrt{\frac{1-r^t}{1-r}},$$

with equality for $g_s \propto (\beta_2/\beta_1)^s$. Hence $\|\Delta\theta_g\|_\infty \le \eta m_g S_T$ after T steps, where $S_T = \sum_{t \le T} C_t$.

Sketch. Apply Cauchy–Schwarz to $m_t = (1-\beta_1)\sum_s \beta_1^{t-s} g_s$, splitting $\beta_1^{t-s} = (\beta_1/\sqrt{\beta_2})^{t-s}\sqrt{\beta_2}^{\,t-s}$. For $(\beta_1, \beta_2) = (0.9, 0.999)$, $C_{100} = 2.24$ and $C_\infty = 7.27$, not the folklore value 1. A geometric rise of the gradients, as at a saddle escape, exceeds 1.

Numerical checks and vacuity. [Numerical]

- *A priori bounds are vacuous.* With no smoothness, the worst-case gradient-norm bound blows up after 5×10^{-5} SGD steps (Remark B1.5). For Adam, Lemma B.4 allows $\|\Delta W_1\|_F \le 8.9 \times 10^3$ after the study's 20000 steps, against a measured 4–11.
- *Adam's ratio.* Real Adam runs reach a per-coordinate ratio $|\hat{m}|/\sqrt{\hat{v}}$ of 1.5–1.7 during the first escape: 97% and 94% of C_{50} at step 50 (x_1, MAJ_3).
- *Condition (D).* In a full-batch GD run at the study's rate (MAJ_3, $n = 2048$, $\eta = 0.2$), (D) holds only for the first 20 steps ($c_{20} = 0.26$). At $\eta = 0.02$ it holds throughout, with $c \ge 0.993$.
- *The chain is loose.* For SGD at step 5000, $\|\Delta\|_{M^{-1}} = 3.7$, $\Lambda = 85$ and $\sqrt{tE} = 159$.
- *Plateaus* (Heuristic). Time-only bounds degrade without limit as the initialization scale $\alpha \to 0$, because plateau time grows while the learned function does not change.

B.3 Why norm-based uniform convergence is vacuous here

Let $\varphi_\gamma(u) = \min(1, \max(0, 1 - u/\gamma))$ be the ramp loss, and $\hat{R}_\gamma(\theta) = \frac{1}{n}\sum_i \varphi_\gamma(y_i f_\theta(x_i))$.

Proposition B.5 (Margin bound and its vacuity criterion; Theorem B2.1 and Lemma B2.1a)**.** *[Proved] Let $C(\theta) \geq 0$ satisfy $\mathfrak{R}_n(\{f_\theta : C(\theta) \leq \phi\}) \leq \kappa\phi/\sqrt{n}$ for all ϕ. Fix $q > 1$, $\phi_0 > 0$, $\gamma_{\max}$ and δ. Put $\gamma_k = \gamma_{\max} q^{-k}$, $j(\theta) = \min\{j \geq 0 : \phi_0 q^j \geq C(\theta)\}$ and $\delta_{jk} = \delta/((j+1)(j+2)(k+1)(k+2))$. With probability $\geq 1 - \delta$, for all θ and all $k \geq 0$,*

$$R(\theta) \leq \hat{R}_{\gamma_k}(\theta) + \frac{2\kappa\phi_0 q^{j(\theta)}}{\gamma_k\sqrt{n}} + \sqrt{\frac{\ln(1/\delta_{j(\theta)k})}{2n}}.$$

Let $c(\theta) = 2\kappa\phi_0 q^{j(\theta)}/\sqrt{n}$ and let $\bar{m}_+(\theta) = \frac{1}{n}\sum_i (y_i f_\theta(x_i))_+$ be the mean positive train margin. If $c(\theta) \geq \max(1, \bar{m}_+(\theta))$, then the right-hand side is at least 1 for every k, whatever $\gamma_{\max}$ is.

Sketch. The first part is the margin bound of Koltchinskii–Panchenko (Mohri et al., Theorem 5.8), with a union bound over the two grids. For the second, $\varphi_\gamma(u) \geq 1 - u_+/\gamma$ gives $\hat{R}_\gamma \geq 1 - \bar{m}_+/\gamma$. Then split into the cases $\gamma_k \leq 1$ and $\gamma_k > 1$.

The capacity constants are proved in the derivation for networks of any widths, with biases included.

- *Path norm* (Proposition B2.2). For $\phi_1(\theta) = \sum_k |a_k|(\sum_j |W_{2,kj}|(\sum_i |W_{1,ji}| + |b_{1,j}|) + |b_{2,k}|)$, every $S \subset [-1, 1]^d$ has $\hat{\mathfrak{R}}_S(\{\phi_1 \leq \phi\}) \leq \phi\sqrt{2\ln(8(d+1))/n}$, i.e. $\kappa_{\text{path}} = 3.32$.
- *Frobenius product* (Proposition B2.3). For $P_F(\theta) = \|a\|(\|[W_1\, b_1]\|_F\|W_2\|_F + \|b_2\|)$, every $S \subset \{\pm 1\}^d$ has $\hat{\mathfrak{R}}_S(\{P_F \leq P\}) \leq \kappa_F P/\sqrt{n}$ with $\kappa_F = \sqrt{31}(1 + \sqrt{4\ln 2}) = 14.84$.

Both proofs rebalance the weights by positive homogeneity, peel the layers with exponential moments and the Ledoux–Talagrand comparison for convex increasing functions (Golowich–Rakhlin–Shamir's device), and finish with Hoeffding's or McDiarmid's inequality. A spectral bound (Bartlett–Foster–Telgarsky, with the initialization as reference matrices) is assembled in Theorem B2.4.

Numerical: vacuous for every network evaluated. [Numerical] The bounds were evaluated on the 49 final Adam networks of the sweep, at $n \in \{256, 2048, 16384\}$. All of them have $\bar{m}_+ = 0.99$–1.01 and $c \geq \max(1, \bar{m}_+)$, so all three bounds are vacuous.

- *Path norm.* The closest case is $c_{\text{path}} = 1.46$, for x_1 at $n = 16384$.
- *Frobenius and spectral.* Over all 49 networks, $c_F = 3.7$–180. The spectral complexity at unit margin is at least 679.
- *The initialization alone is too large.* Already at the initialization's norms, $\phi_1(\theta_0) \approx (2/\pi)^{3/2}\sqrt{Nd} \approx 31.5$ and $P_F(\theta_0) \approx \sqrt{N} \approx 11.3$. This forces $c \geq 1$ for $n \lesssim 4.4 \times 10^4$ (path norm) and $n \lesssim 1.1 \times 10^5$ (Frobenius).
- *A priori version.* Combined with Theorem B.1, the Frobenius bound (Theorem B3.1) is vacuous even at $t = 0$: $c_F = 2.8$ at $n = 16384$ and 22 at $n = 256$.

Heuristic reading: the norms are dominated by the random features present at initialization. It is not a theorem, and training can lower the norms: the path norm of x_1 at $n = 16384$ is 26 after training and 32 at initialization.

B.4 PAC-Bayes centred at the initialization

Let Σ be a finite grid of per-group noise levels $\sigma = (\sigma_1, \sigma_2, \sigma_3)$, with $|\Sigma| = G$. Put $Q_{\theta,\sigma} = \mathcal{N}(\theta, \mathrm{diag}(\sigma_g^2 I_{p_g}))$ and $P_\sigma = \mathcal{N}(\theta_0, \mathrm{diag}(\sigma_g^2 I_{p_g}))$, and let $\mathrm{KL} = \sum_g \|\theta_g - \theta_{0,g}\|^2/2\sigma_g^2$ be their relative entropy.

Theorem B.6 (PAC-Bayes with the prior at initialization; Theorem B3.2)**.** *[Proved] Let $n \geq 8$. With probability $\geq 1 - \delta_0$, for all θ (however obtained) and all $\sigma \in \Sigma$,*

$$R(Q_{\theta,\sigma}) \leq \mathrm{kl}_+^{-1}\Big(\hat{R}_S(Q_{\theta,\sigma}),\ \tfrac{1}{n}\big[\mathrm{KL} + \ln \tfrac{2G\sqrt{n}}{\delta_0}\big]\Big). \tag{PB}$$

Derandomized form. *Suppose θ and σ are fixed first, by any rule. Then draw, independently of that choice, m iid pairs (θ'_j, B_j) with $\theta'_j \sim Q_{\theta,\sigma}$ and B_j a batch drawn from S. Let $\hat{e}$ be the mean batch error, and $\hat{\Delta}$ the mean sign disagreement of $f_{\theta'_j}$ and f_θ on m fresh batches of unlabelled uniform inputs. With probability $\geq 1 - \delta_0 - \delta_1 - \delta_2$,*

$$R(\theta) \leq \mathrm{kl}_+^{-1}\Big(\mathrm{kl}_+^{-1}(\hat{e}, \tfrac{\ln(1/\delta_1)}{m}),\ \tfrac{1}{n}\big[\mathrm{KL} + \ln \tfrac{2G\sqrt{n}}{\delta_0}\big]\Big) + \mathrm{kl}_+^{-1}(\hat{\Delta}, \tfrac{\ln(1/\delta_2)}{m}).$$

Sketch. The prior depends only on θ_0, which is independent of S. So the PAC-Bayes-kl bound (Maurer 2004, Theorem 5; in the line of [21]) applies conditionally on θ_0, with a union bound over Σ. The Monte Carlo terms use kl-Chernoff on the fresh draws. Derandomization uses $\nVdash[yf_\theta \leq 0] \leq \nVdash[yf_{\theta'} \leq 0] + \nVdash[\mathrm{sgn} f_{\theta'} \neq \mathrm{sgn} f_\theta]$. The approach is that of [13], with the prior centred at the initialization.

Theorem B.7 (Time-based KL: "short training implies generalization"; Theorem B3.3)**.** *[Proved] Under the hypotheses of Theorem B.6, with $\lambda_\sigma = \max_g m_g/2\sigma_g^2$, the KL in (PB) at the trained parameters satisfies:*

- *(a) for gradient flow, or GD under (D) with constant c ($c = 1$ for the flow): $\mathrm{KL} \leq t(L_0 - L_S(\theta(t)))\lambda_\sigma/c \leq tL_0\lambda_\sigma/c$, which is linear in time;*
- *(b) for any update: $\mathrm{KL} \leq t_K \sum_g m_g E_{K,g}/2\sigma_g^2$, with the measured energies of Corollary B.2(a);*
- *(c) for Adam: $\mathrm{KL} \leq \sum_g p_g(\eta m_g S_K)^2/2\sigma_g^2$, which is quadratic in K, since $S_K \geq K$;*
- *(d) for SGD, under the hypotheses of Proposition B.3, with $V(t) = 4tL_0 + 2\eta t\nu^2(1 + \beta t)/B$: with probability $\geq 1 - \delta_0 - \delta'$ over S and the minibatches, (PB) holds at θ_K with $\mathrm{KL} \leq \lambda_\sigma V(t)/\delta'$; and the randomized predictor "rerun SGD, then perturb" satisfies the PAC-Bayes-kl inequality with $\mathrm{KL} \leq \lambda_\sigma V(t)$.*

Sketch. $\mathrm{KL} \leq \lambda_\sigma \|\theta - \theta_0\|^2_{M^{-1}}$. Insert Theorem B.1, Corollary B.2 or Lemma B.4. For (d), insert Proposition B.3 together with Markov's inequality, or with convexity of KL for the mixture.

Numerical. [Numerical]

- *Final Adam networks* (20000 steps). For the deterministic network, the bound is 0.26–0.56 at $n = 16384$ for the five targets with at most three relevant variables (x_1, AND_3, MAJ_3, x_1x_2, $x_1x_2x_3$). It is 0.55–1.0 at $n = 2048$, and vacuous at $n = 256$. For targets with 4–6 relevant variables it is uninformative even at $n = 16384$ (0.79–1.0; KL 3500–6000 nats). For x_1 at $n = 16384$ the KL is 1804 nats, against about 4 nats for describing x_1 in any circuit language.

- *Early checkpoints.* After 100–300 steps of SGD or Adam at $n = 16384$ the bounds are 0.10–0.34. With a union bound over the 10 checkpoints evaluated per target, the best are 0.11 (x_1) and 0.23 (MAJ_3).
- *Time-based forms.* These are informative (< 0.5) only up to gradient-flow time $t \approx 4$, or 50 Adam steps.
  - (a), GD at $\eta = 0.02$: 0.59–0.77 over steps 200–1000.
  - (b): 0.39 for SGD on x_1 at step 20, then at least 0.9 from step 100 on. (b) is a diagnostic, always weaker than (PB) with the measured displacement; its KL is 2–44 times the flow value.
  - (c), Adam: 0.44–0.56 at step 50, and vacuous by step 200.
  - (d) is not evaluated, because smoothness fails for ReLU.

B.5 Distillation certificates without labelled hold-out data

Setting. $\{\mathcal{H}_c\}_{c\in I}$ is a countable family of finite classes of predictors $\{\pm1\}^d \to \{\pm1\}$, with weights $\pi(c) > 0$, $\sum_c \pi(c) \le 1$, all fixed before seeing data. Training randomness ξ is independent of S. The predictors to be certified, $h_1, \dots, h_J$ (for example checkpoints), are arbitrary functions of (S, ξ). $U = (u_1, \dots, u_m)$ is iid from the input distribution and independent of (S, ξ).

Theorem B.8 (Distillation certificate; Theorem C1)**.** *[Proved] With probability $\ge 1 - \delta_1 - \delta_2$ over (S, ξ, U), simultaneously for every $j \le J$, every $c \in I$ and every $g \in \mathcal{H}_c$ (in particular for a g selected using S, ξ, h_j and U),*

$$\mathrm{err}(h_j) \le \mathrm{kl}_+^{-1}(\widehat{\mathrm{err}}_S(g), \varepsilon_c^S) + \mathrm{kl}_+^{-1}(\widehat{\mathrm{dis}}_U(h_j, g), \varepsilon_c^U), \qquad \varepsilon_c^S = \frac{\ln|\mathcal{H}_c| + \ln\frac{1}{\pi(c)} + \ln\frac{1}{\delta_1}}{n}, \quad \varepsilon_c^U = \frac{\ln|\mathcal{H}_c| + \ln\frac{1}{\pi(c)} + \ln\frac{J}{\delta_2}}{m}.$$

(b) If a list of K candidates (c_k, g_k) is chosen from (S, ξ, U_A), and U_B (m_B points) is independent of (S, ξ, U_A), then ε^U may be replaced by $\ln(JK/\delta_2)/m_B$. (c) If $\mathrm{dis}(h_j, g)$ is computed exactly, the second term is $\mathrm{dis}(h_j, g)$, at probability $1 - \delta_1$.

Sketch. The Occam event, a kl-Chernoff and union bound over all of $\bigcup_c \mathcal{H}_c$ [7], involves neither h nor the selection rule. So g may be fitted to h, which was fitted to S. Conditionally on (S, ξ), the h_j are fixed and U is still iid. The triangle inequality $\mathrm{err}(h) \le \mathrm{err}(g) + \mathrm{dis}(h, g)$ combines the two. The disagreement must not be estimated on the training inputs, and the network must not be trained on U.

The classes, counted. [Proved]

- *Fourier signs* (Proposition C1.2). Let $\mathcal{F}_{k,T} = \{\mathrm{sgn}(w_0 + \sum_{j\le T} w_j \chi_{S_j}) : S_j \text{ distinct}, \ 1 \le |S_j| \le k\}$, where $\chi_S(x) = \prod_{i\in S} x_i$. For a fixed support the class has VC dimension exactly $T + 1$, and

  $$\ln|\mathcal{F}_{k,T}| \le \ln\tbinom{D_k^*}{T} + \ln\Phi(2^{\min(T,d)}, T), \qquad \Phi(N, T) = 2\textstyle\sum_{i\le T}\tbinom{N-1}{i},$$

  where $D_k^* = \sum_{1\le i\le k}\binom{d}{i}$. The proof: on the cube the characters of a support take 2^{rank} values, with the rank over GF(2), and Cover's count bounds the threshold functions on that point set.
- *Fan-in-2 circuits* (Proposition C1.3). The functions computed by circuits of at most T gates over all 16 binary gates number at most $\sum_{s\le T}(16(d+s)^2)^s(d+s)/s!$. The proof: gate renaming acts freely on reduced, duplicate-free descriptions.

- *Juntas* (Proposition C1.4), counted exactly: $|\mathcal{J}_{\le k}| = \sum_{j\le k}\binom{d}{j}N_j$ with $N_j = \sum_{i\le j}(-1)^{j-i}\binom{j}{i}2^{2^i}$, by inclusion–exclusion.

At $d = 30$ the sizes in nats are as follows (`theory/checks/c1_class_sizes.txt`):

size parameter (k or T)	1	2	3	4	6	8
$\ln\lvert\mathcal{J}_{\le k}\rvert$	4.13	8.39	13.70	21.29	57.66	–
$\ln\lvert\mathcal{F}_{1,T}\rvert$	4.8	8.7	13.2	18.5	32.1	49.9
$\ln\lvert\mathcal{C}_{\le T}\rvert$	13.1	22.2	31.0	39.6	56.6	73.4

Numerical. [Numerical] Juntas and Fourier signs were evaluated on the saved final Adam networks (13 targets, 3 seeds; `theory/checks/c1_fourier_cert.txt`). The selection used h and unlabelled inputs only, with $K = 169$ candidates and $\delta = 0.05$; circuits were not searched.

- *Coverage.* At $n = 2048$ every target is certified below 0.5 for every seed, and below 0.04 for 9 of the 13. At $n = 16384$ every network is certified at ≤ 0.021.
- *Threshold-like targets.* The Fourier-sign class certifies MAJ_9 at 0.034 and the 10-variable threshold at 0.09–0.10 ($n = 2048$) and 0.02 ($n = 16384$). Juntas fail on these targets.
- *Soundness check.* No certificate falls below the true error; the smallest gap is 0.0009. The certificates are for the networks rebuilt from float16 parameters, and transferring them to float32 costs +0.0024.
- *Along training.* The class that minimizes the certificate grows in discrete stages. For example, for stair_5 it is x_1 alone (5 nats) during steps 50–200, and a 5-junta (34 nats) from step 500 on.

Weakness: the certificate is a posteriori. It does not explain why gradient descent produces a network close to a small circuit.

B.6 One-pass training: online-to-batch conversions

In the online protocol each step draws a fresh batch, the current predictor predicts on it, and then the network updates. Over a run, N samples are drawn in total. Sample j is predicted by $h_{(j)}$, which depends only on the past. Let $\ell_j = \mathbb{1}[h_{(j)}(x_j) \ne y_j]$, $p_j = \mathrm{err}(h_{(j)})$ and $M_N = \sum_{j\le N}\ell_j$.

Proposition B.9 (Progressive validation; Proposition B3.4)**.** *[Proved] Let step k draw a fresh batch of B iid examples, and let e_k be the error of the pre-update parameters θ_{k-1} on it. For any window of K' steps fixed in advance, with probability $\ge 1-\delta$, $\frac{1}{K'}\sum_{k\in\mathrm{window}} R(\theta_{k-1}) \le \frac{1}{K'}\sum_{k\in\mathrm{window}} e_k + \sqrt{\ln(1/\delta)/(2K'B)}$.*

Proposition B.10 (Online-to-batch for the trajectory; Proposition C4.1)**.** *[Proved] With probability $\ge 1-\delta$, $\frac{1}{N}\sum_j p_j \le \frac{M_N}{N} + \sqrt{\frac{\ln(1/\delta)}{2N}}$. For any finite grid $\Lambda \subset (0,\infty)$, with probability $\ge 1-\delta$, simultaneously for all $\lambda \in \Lambda$,*

$$\frac{1}{N}\sum_{j\le N} p_j \le \frac{\lambda M_N + \ln(|\Lambda|/\delta)}{(1-e^{-\lambda})N}.$$

Proposition B.11 (Best checkpoint and final iterate; Proposition C4.2)**.** *[Proved] Let I be a finite set of steps, and let $\widehat{\mathrm{err}}_{\ge i}(h_i)$ be the error of h_i on the $N_{\ge i}$ samples drawn at steps $\ge i$. (1) With probability $\ge 1-\delta$, $\mathrm{err}(h_i) \le \mathrm{kl}_+^{-1}(\widehat{\mathrm{err}}_{\ge i}(h_i), \ln(|I|/\delta)/N_{\ge i})$ for all $i \in I$. (2) For m unlabelled inputs U independent of the run, with probability $\ge 1-\delta-\delta'$, $\mathrm{err}(h_{\mathrm{final}})$ is at most*

$$\min_{i\in I}\left[\mathrm{kl}_+^{-1}\left(\widehat{\mathrm{err}}_{\ge i}(h_i), \tfrac{\ln(|I|/\delta)}{N_{\ge i}}\right) + \mathrm{kl}_+^{-1}\left(\widehat{\mathrm{dis}}_U(h_{\mathrm{final}}, h_i), \tfrac{\ln(|I|/\delta')}{m}\right)\right].$$

(3) A small trajectory average alone implies nothing about the final iterate.

Sketch. Each example is fresh for the predictor that sees it first, so $p_j - \ell_j$ are martingale differences, and Hoeffding's lemma gives the first two statements. The fast rate uses the supermartingale $\exp(\sum_j [(1-e^{-\lambda})p_j - \lambda \ell_j])$. For Proposition B.11, the samples of later steps are fresh for h_i. This is the classical online-to-batch principle [9].

Numerical. [Numerical] The bounds were evaluated on one-pass Adam runs with logged progressive errors.

- *Proposition B.9.* The bound is within 0.006 of the measured mean test error of the window's iterates in every window fixed in advance (slack 0.0057).
- *Proposition B.10, Hoeffding form.* On 18 online reruns it certifies the average test error over the second half of training at ≤ 0.0017, and at ≤ 0.0038 over the last 2000 steps (`study/ONLINE-BOUNDS.md`).
- *Not evaluated.* The fast-rate form, and Proposition B.11, whose later-sample errors are not logged.

These bounds concern the random-iterate predictor, not the final network.

B.7 Two-layer ReLU networks from balanced small initialization

Setting (Part I of D). Put $\tilde{x} = (x, 1) \in \mathbb{R}^{d+1}$, $f(x) = \sum_{j \leq m} a_j \sigma(w_j \cdot \tilde{x})$, $L = \frac{1}{2}\|y - f\|_n^2$ and $r = y - f$.

- *Data constants.* λ_X is the top eigenvalue of $\frac{1}{n}\sum_i \tilde{x}_i \tilde{x}_i^\top$.
- *Utilities.* The utility of a unit direction v for a residual r is $U_r(v) = \langle r, \sigma(v \cdot \tilde{x}) \rangle_n$. Write $U^*(r) = \sup_{\|v\|=1} |U_r(v)|$ and $U_0^* = U^*(y)$.
- *Initialization.* It is balanced, $|a_j(0)| = \|w_j(0)\| = \alpha_j$, with signs ϵ_j. The masses are $\rho_j = a_j^2 + \|w_j\|^2$, and $P = \sum_j \rho_j = \|\theta\|^2$. Width enters only through $P(0)$, which is $2m\alpha^2$ for $\alpha_j = \alpha$.
- *Flow.* A *selection flow* is gradient flow with any measurable choice $s_j(t,i) \in \partial\sigma(w_j \cdot \tilde{x}_i)$ at the kinks. It includes every Clarke subgradient flow.

Lemma B.12 (Exact structure; Lemma D1(b), (c), (g))**.** *[Proved] For every selection flow from a balanced initialization, any data and any width:*

(i) Mass clock. $\rho_j(t) = \rho_j(0) \exp\big(2\int_0^t \epsilon_j U_{r(s)}(\hat{w}_j(s))\, ds\big)$ *with* $\hat{w}_j = w_j/\|w_j\|$, *and* $\operatorname{sgn} a_j(t) = \epsilon_j$.

(ii) Measure form. $f_t(x) = \sum_j \epsilon_j \frac{\rho_j(t)}{2} \sigma(\hat{w}_j(t) \cdot \tilde{x})$ *for every* x, *so the variation norm of* f_t *is at most* $P(t)/2$.

(iii) Merging law. *For* $\epsilon_j = \epsilon_k$, *let* $Q_{jk} = a_j a_k - w_j \cdot w_k = \|w_j\|\|w_k\|(1 - \hat{w}_j \cdot \hat{w}_k) \geq 0$. *Almost everywhere* $\dot{Q}_{jk} = \epsilon_j \|w_j\|\|w_k\| \frac{1}{n}\sum_i r_i m_{jk}(t,i)$, *where* $m_{jk} = (s_j - s_k)(\hat{w}_j - \hat{w}_k) \cdot \tilde{x}_i \geq 0$ *vanishes except at training points that separate* j *and* k. *Hence:*

(g1) Q_{jk} *is constant while* j *and* k *use the same selections at every training point;*

(g2) Q_{jk} *is nonincreasing while* $\epsilon_j r_i \leq 0$ *at every separating point ("attracting" points);*

(g3) in general, $\frac{d}{dt}\sqrt{Q_{jk}} \leq \sqrt{(d+1)\|w_j\|\|w_k\|/2}\; \bar{r}_{jk}^{\mathrm{rep}}$, *where* $\bar{r}_{jk}^{\mathrm{rep}}$ *is the residual mass* $\frac{1}{n}\sum |r_i|$ *over separating points with* $\epsilon_j r_i > 0$.

Sketch and credit. The derivative of $a_j^2 - \|w_j\|^2$ vanishes because $s\,z = \sigma(z)$ for every $s \in \partial\sigma(z)$. So $|a_j| = \|w_j\|$, and positive homogeneity turns $\dot\rho_j = 4a_j\langle r, \sigma(w_j \cdot \tilde x)\rangle_n$ into $2\epsilon_j\rho_j U_r(\hat w_j)$. For (iii), $\dot Q_{jk} = (\xi_j - \xi_k)\cdot(a_k w_j - a_j w_k)$, with $\xi_j = \frac{1}{n}\sum_i r_i s_j \tilde x_i$, and $\partial\sigma$ is monotone. (g1) is not new. It is the off-diagonal entry of the classical balancedness law $W_1 W_1^\top - W_2^\top W_2 = \text{const}$ of two-layer linear networks [4, 12], applied to a pair of neurons that share their selections and so form, locally, a linear network. For ReLU networks only the diagonal entries are conserved globally (Marcotte–Gribonval–Peyré, NeurIPS 2023). Only the monotonicity statements (g2) and (g3) may be new. While (g1) or (g2) holds, $1 - \cos\angle(w_j, w_k)$ shrinks like 1/mass.

Theorem B.13 (No growth before $\ln(1/\alpha)/U_0^*$; Theorem D2)**.** *[Proved] For any selection flow from a balanced initialization, any data and any width:*

(1) $\rho_j(t) \le \rho_j(0)\exp\big(2\int_0^t U^*(r(s))ds\big)$*, and the same for* P*;*

(2) $\|f_t\|_n \le \frac{\sqrt{\lambda_X}}{2}P(t)$ *and* $\sup_{x\in\{\pm1\}^d}|f_t(x)| \le \frac{\sqrt{d+1}}{2}P(t)$*;*

(3) for $\beta > P(0)$*, the escape time* $t_\beta = \inf\{t : P(t) \ge \beta\}$ *satisfies*

$$t_\beta \ge T_\beta := \frac{\ln(\beta/P(0))}{2(U_0^* + \lambda_X\beta/2)}.$$

For $t \le T_\beta$ *the network is the empty circuit up to* $\sup_x|f_t(x)| \le \frac{\sqrt{d+1}}{2}\beta$*, and* $L(t) \ge \frac{1}{2}(\|y\|_n - \frac{\sqrt{\lambda_X}}{2}\beta)_+^2$*;*

(6) the leading constant is sharp. Every selection flow with $\alpha_j = \alpha$ *has* $\liminf_{\alpha\to0} t_\beta U_0^*/\ln(1/\alpha) \ge 1$*, at every fixed* $\beta > 0$ *and width. Equality holds in an explicit instance:* $y = x_1$ *on the full cube, one neuron started at* $v^* = (e_1 + e_0)/\sqrt{2}$ *with* $\alpha < 2^{-1/4}$*, and selection* 0 *on the kink set* $\{x_1 = -1\}$*. There* $U_0^* = 1/\sqrt{2}$*,* $P(t)$ *is logistic, and* $t_\beta = \frac{\ln(\beta/P(0))}{2U_0^*} + \frac{1}{\sqrt{2}}\ln\frac{\sqrt{2}-P(0)}{\sqrt{2}-\beta}$ *for* $P(0) < \beta < \sqrt{2}$*.*

Sketch. (1) follows from the mass clock, and (2) from the measure form. For $t < t_\beta$, $U^*(r_t) \le U_0^* + \sqrt{\lambda_X}\|f_t\|_n < U_0^* + \lambda_X\beta/2$, and (1) then gives (3). For the lim inf in (6), use $t_\beta \ge t_{\beta'} \ge T_{\beta'}$ for $\beta' \in (P(0), \beta]$, let $\alpha \to 0$, and then let $\beta' \to 0$. The derivation also gives a discrete-GD form of (1) (item 5), and a certified upper bound $\hat U(r) \ge U^*(r)$ from two eigenvalue computations (Lemma D2.1), which may replace U^* in (1) and U_0^* in (3). The lower-order terms (the $\ln m$ term, the β-factor) are not claimed sharp. That the escape actually happens near T_β is not proved here.

Theorem B.14 (Path norm through the escape; Theorem D3)**.** *[Proved] For every* $\beta > P(0)$ *and* $t \ge 0$*: (1) a priori,* $P(t) \le \beta + \|y\|_n^2(t - T_\beta)_+$*; (2) from the training-loss curve,* $\sqrt{P(t)} \le \sqrt{\beta} + \int_{T_\beta}^t \sqrt{-\dot L(s)}\,ds$*.*

Sketch. $\dot P = 4\langle r, f\rangle_n \le \|y\|_n^2$, and $\frac{d}{dt}\|\theta\| \le \|\dot\theta\| = \sqrt{-\dot L}$. At small initialization $P \ge 2\|\mu_t\|_{TV}$ measures the function, not random features.

Theorem B.15 (Width-free generalization through the escape; Theorem D3.2)**.** *[Proved] Fix, before seeing data:*

- *for* $K \ge 1$*, a net* $\mathcal{N}_{\eta_K}$ *of the unit sphere* S^d *with* $\eta_K = 1/\sqrt{K(d+1)}$ *and* $|\mathcal{N}_\eta| \le (1 + 2/\eta)^{d+1}$*;*
- *for dyadic* $B = 2^\ell$*, the classes* $\mathcal{G}_{K,B} = \{\frac{B}{K}\sum_{k\le K}\epsilon_k\sigma(u_k\cdot\tilde x) : \epsilon_k \in \{-1, 0, 1\}, u_k \in \mathcal{N}_{\eta_K}\}$*;*
- *a finite margin grid* Γ*.*

Let the inputs be uniform, and put $b^2 = \frac{1}{2}(1+\lambda_X)$ *and* $e_K = 2B^2(b+1)^2/(K\gamma^2)$*. With probability* $\geq 1-\delta$*, the following holds simultaneously for every two-layer ReLU network* $f = \sum_j a_j \sigma(w_j \cdot \tilde{x})$ *of any width, balanced or not (so for every* f_t *and every GD iterate), every* $K \geq 1$*, every* $\gamma \in \Gamma$ *and every dyadic* $B \geq \sum_j |a_j| \|w_j\|$*:*

$$\operatorname{err}(\operatorname{sgn} f) \leq \operatorname{kl}_+^{-1}\Big(\widehat{P}_S[yf \leq 2\gamma] + e_K,\ \tfrac{K\ln(2|\mathcal{N}_{\eta_K}|+1)+\ln\frac{2|\Gamma|K(K+1)(|\ell|+1)(|\ell|+2)}{\delta}}{n}\Big) + e_K.$$

Along a selection flow, B *may be taken from Theorem B.14, since* $\sum_j |a_j| \|w_j\| \leq P/2$*.*

Sketch. Sample K neurons from the measure of Lemma B.12(ii) (Maurey's method), with weights $|a_j|\|w_j\|$. This gives an $L^2(\nu)$ error $\leq Bb/\sqrt{K}$ for $\nu = \frac{1}{2}(P_X + \widehat{P}_{S,X})$. Rounding the directions to the net costs $B/\sqrt{K}$ in sup norm. Chebyshev's inequality under both the test and the training distribution turns the L^2 error into the margin slack e_K, and kl-Occam applies over the finite classes. *Vacuous at realistic sizes:* at $d = 16$ the bound is non-vacuous only for $n \gtrsim 10^5$, and it is 1.0 at every checkpoint of the runs below, both with the measured P and with the a priori bound.

The derivation also proves a *K-neuron Occam bound* (Proposition D3.3, [Proved]). With probability $\geq 1-\delta$, simultaneously for every predictor f, every K, every dyadic B, every $\gamma \in \Gamma$, and every $g = \sum_{k \leq K} c_k \sigma(v_k \cdot \tilde{x})$ with unit v_k, $\sum_k |c_k| \leq B$ and $\sup_{\{\pm 1\}^d} |f - g| \leq \gamma/4$:

$$\operatorname{err}(\operatorname{sgn} f) \leq \operatorname{kl}_+^{-1}(\widehat{P}_S[yf \leq \gamma],\ [C'_{K,B,\gamma} + \ln\tfrac{2|\Gamma|(K+1)(K+2)(|\ell|+1)(|\ell|+2)}{\delta}]/n),$$

$$C'_{K,B,\gamma} = K\big[(d+1)\ln(1+\tfrac{16B\sqrt{d+1}}{\gamma}) + \ln(2\lceil\tfrac{8KB\sqrt{d+1}}{\gamma}\rceil + 1)\big].$$

It is a posteriori in the clusters. The next proposition says when the network stays close to such a g.

Proposition B.16 (Persistence of clusters; Proposition D4.1)**.** *[Conditional] Fix a checkpoint* t_1*, a partition of the neurons into same-sign clusters* $G_1, \dots, G_K$ *and a remainder* D*, and reference neurons* $j_k \in G_k$*. Suppose that on* $[t_1, t]$*:*

(A) for every k*, every* $j \in G_k$ *and almost every* s*, every training point separating* j *and* j_k *satisfies* $\epsilon_j r_i(s) \leq 0$*.*

Let $g_t = \sum_k c_k(t)\sigma(\hat{w}_{j_k}(t) \cdot \tilde{x})$ *with* $c_k = \sum_{j \in G_k} \epsilon_j \rho_j(t)/2$*, and let* $P_D = \sum_{j \in D} \rho_j$*. Then* $\sum_k |c_k| \leq P(t)/2$ *and*

$$\sup_x |f_t(x) - g_t(x)| \leq \sqrt{d+1}\Big[\sqrt{2}\sum_k \sum_{j \in G_k} \frac{\|w_j(t)\|^{3/2}}{\|w_{j_k}(t)\|^{1/2}}\sqrt{Q_{jj_k}(t_1)} + \tfrac{1}{2}P_D(t_1)\, e^{2\int_{t_1}^t \hat{U}(r_s)ds}\Big].$$

If (A) fails, the same holds with $\sqrt{Q_{jj_k}(t_1)}$ *replaced by* $\sqrt{Q_{jj_k}(t_1)} + \int_{t_1}^t \sqrt{(d+1)\|w_j\|\|w_{j_k}\|/2}\,\bar{r}^{\mathrm{rep}}_{jj_k}\,ds$*. Whenever the right-hand side is at most* $\gamma/4$*, Proposition D3.3 applies at time* t *with this* K*.*

Sketch. Under (A), Q_{jj_k} is nonincreasing by (g2), and $\frac{\rho_j}{2}\|\hat{w}_j - \hat{w}_{j_k}\| = \|w_j\|^2\sqrt{2Q_{jj_k}/(\|w_j\|\|w_{j_k}\|)}$. The dormant neurons are bounded by the clock (Theorem B.13(1)) from t_1 on. Condition (A) is checkable from the training residuals along the run, but it is not proved a priori.

Conjecture B.17 (A priori early-phase theorem; Conjecture D4)**.** *[Conjecture] Fix* S *and the width, take initial directions in general position, and let* $\alpha \to 0$*. Let* K *be the number of distinct selection-critical directions of* $\pm U_y$ *that attract a positive fraction of the initial directions under the direction flow. Suppose that the* K*-neuron least-squares fit on these directions has residual* $r^\star$ *with* $\epsilon r_i^\star < 0$

at every training point on the clusters' boundary hyperplanes. Then for every $c < c^\star(S)$, where $c^\star \ln(1/\alpha)/U_0^*$ is the time to the next escape,

$$\sup_{t \le (1+c)\ln(1/\alpha)/U_0^*} \inf_g \sup_x |f_t - g| \to 0$$

over K-neuron functions g. Proposition D3.3 would then apply uniformly over that window, with K computable from S alone.

Numerical. [Numerical] Setup: $d = 16$, $n = 1000$–4000, width 64, $\alpha = 10^{-6}$–10^{-2}, small-step GD as a proxy for the flow, and exact test error over all 2^{16} inputs.

- *Clock tightness.* The growth bound reaches 0.84–0.95 of its exponent. At $\alpha \le 10^{-3}$, T_β is 60–81% of the observed escape time.
- *Condensation.* At $\alpha \le 10^{-3}$ the neurons condense into $K = 2$ clusters with $1 - \cos = 2 \cdot 10^{-8}$–$2 \cdot 10^{-4}$. In the $n = 4000$ run, one of the 64 neurons stays outside them at small mass.
- *Clusters sit on kinks.* Except for x_1, the clusters sit on data hyperplanes, with training margins 10^{-12}–$3 \cdot 10^{-7}$. So arguments that freeze activation patterns cannot work.
- *Persistence.* On intervals where (A) holds, Q_{jk} increased by at most 0.2%, which supports Conjecture B.17.
- *Certificates.* The K-neuron bound with the network's own two clusters certifies MAJ_3 at 0.19 ($n = 1000$) and 0.051 ($n = 4000$), where the true error is 0. These values are for the balanced surrogate of the GD network; for one configuration the referee recomputed the value on the GD network itself and got the same number.
- *The race model's reservoir can be empty.* On $x_1 + \frac{1}{2}x_2x_3$ at $\alpha = 10^{-6}$, every neuron is recruited at the first escape, and the network stalls at a 2-neuron point up to $t = 400$, never learning x_2x_3. This is the "two-edged sword" of early alignment [8].

B.8 Quadratic networks with fixed output signs

Setting (Part II of D). For $U \in \mathbb{R}^{d\times h}$ and $V \in \mathbb{R}^{d\times h'}$, the network is

$$f_{U,V}(x) = \sum_j [(u_j \cdot x)^2 - \|u_j\|^2] - \sum_j [(v_j \cdot x)^2 - \|v_j\|^2] = \langle X, xx^\top - I\rangle, \qquad X = UU^\top - VV^\top.$$

- *The model.* The output weights are fixed at ± 1 and the units are centred, so this is symmetric matrix sensing with measurements $A_i = x_i x_i^\top - I$. Trainable output weights, which make the dynamics 3-homogeneous, are not covered.
- *Gradient flow.* With $M_r = \frac{1}{n}\sum_i r_i A_i$, the flow is $\dot U = 2M_rU$ and $\dot V = -2M_rV$.
- *Initialization.* $U(0) = \alpha U_0$ and $V(0) = \alpha V_0$, with $s_0 = \max(\|U_0\|_{op}, \|V_0\|_{op})$.
- *Grown count.* $N_\epsilon(t)$ is the number of eigenvalues of $UU^\top$ and of $VV^\top$ that are at least ϵ.
- *Clock unit and effective time.* $\Theta_\epsilon = \frac{1}{4}\ln\frac{\epsilon}{\alpha^2 s_0^2} \approx \frac{1}{2}\ln\frac{1}{\alpha}$ and $\tau(t) = \int_0^t \|r\|_n$.
- *Coherence.* $\kappa = \sup_{Z=Z^\top\neq 0} \frac{1}{n}\sum_i \langle Z, A_i\rangle^2/\|Z\|_F^2 = 2\lambda_{\max}(\frac{1}{n}\Phi^\top\Phi)$, where $\Phi_{i,(k,l)} = x_{ik}x_{il}$ for $k < l$. It depends only on the training inputs, not on labels, trajectory or width, and its population value is 2.

- *Path norm.* $P(t) = \|U\|_F^2 + \|V\|_F^2$.

Lemma B.18 (Log-det clock; Theorem D5)**.** *[Proved] If $\dot{W} = A(t)W$ with A continuous and symmetric, then for every $K \le \min(d, h)$*

$$\sum_{k\le K} \ln \sigma_k(W(t)) \le \sum_{k\le K} \ln \sigma_k(W(0)) + \int_0^t \sum_{k\le K} \lambda_k(A(s))\, ds.$$

For $W_{j+1} = (I + \eta A_j)W_j$ with $I + \eta A_j \succ 0$, the same holds with $\sum_j \sum_{k\le K} \ln(1 + \eta\lambda_k(A_j))$ in place of the integral.

Sketch. Let E hold the top-K right singular vectors of $W(t)$, and put $G = (WE)^\top WE$. Then $\frac{d}{ds} \ln\det G = 2\,\mathrm{tr}(\Pi_s A) \le 2\sum_{k\le K} \lambda_k(A)$ by Ky Fan's maximum principle, and interlacing handles $s = 0$. The discrete case follows from Horn's inequality.

Theorem B.19 (Time limits rank; Theorem D6)**.** *[Proved] For gradient flow, every t and every ϵ with $\Theta_\epsilon > 0$:*

(1) Clock form. *$N^U \Theta_\epsilon \le \int_0^t \sum_{k\le N^U} \lambda_k(M_{r(s)})ds$, and likewise for V with $-M_r$, where $N^U + N^V = N_\epsilon(t)$.*

(2) Rank bound. *$N_\epsilon(t) \le \kappa\,(\tau(t)/\Theta_\epsilon)^2$.*

(3) No early growth. *The first time t_ϵ at which some eigenvalue reaches ϵ satisfies $t_\epsilon \ge \Theta_\epsilon/(U_0^* + 2\kappa\sqrt{d}\,\epsilon)$, where $U_0^* = \|M_y\|_{op}$.*

(4) Path norm. *$P(t) \ge \|X(t)\|_*$, and $\dot{P} = 4\langle r, f\rangle_n \le \|y\|_n^2$.*

(5) GD and SGD. *Let M_j be the full-batch (GD) or minibatch (SGD) matrix of step j. Assume the step-size condition $2\eta\|M_j\|_{op} < 1$ at every step.*

- GD. *Items 1–3 hold with $\int_0^t \to \eta\sum_{j<J}$, $\tau_J = \eta\sum_{j<J}\|r_j\|_n$ and $t_\epsilon = \eta J_\epsilon$.*
- SGD. *Item 1 holds with the minibatch matrices. Item 2 becomes $N \le \big(\eta\sum_j \sqrt{\kappa_{B_j}}\|r_j\|_{B_j}\big)^2/\Theta_\epsilon^2$, with κ_{B_j} the coherence of batch B_j. Item 3 holds with $\max_{j<J_\epsilon}\|M_y^{B_j}\|_{op}$ and $\max_{j<J_\epsilon}\kappa_{B_j}$ in place of U_0^* and κ. This SGD form of item 3 was added after the second referee pass and has not been refereed.*
- Path norm, with no step-size condition. *$P_J \le \exp\big(4\eta^2\sum_{j<J}\|M_j\|_{op}^2\big)\big(P(0) + \eta\sum_{j<J}\|y\|_{B_j}^2\big)$. The flow bound of item 4 can fail for GD.*

Sketch. Item 1 is Lemma B.18 with $A = \pm 2M_r$, together with $\sigma_k(U(t))^2 \ge \epsilon$ and $\sigma_k(U(0)) \le \alpha s_0$. For item 2, let $Z = \Pi_+ - \Pi_-$ project onto the relevant eigenvectors of M_r, so that $\|Z\|_F^2 \le N$. Then the two Ky Fan sums are at most $\langle M_r, Z\rangle = \langle r, \mathcal{A}(Z)\rangle_n \le \|r\|_n\sqrt{\kappa N}$, so $N\Theta_\epsilon \le \tau\sqrt{\kappa N}$. The bound is *weight-free but not a priori*: τ is read off the training-loss curve. The time-only form $\tau \le R_0 t$, with $R_0 = \|r(0)\|_n$, is much weaker; for x_1x_2 below it gives $N \le 42$ at $t = 20$, against 4 from τ. For diagonal measurements and diagonal initialization the theorem gives a slightly weaker form of the diagonal-network rank bound (Theorem C2.3).

Theorem B.20 (Generalization uniform in time; Theorem D7)**.** *[Proved] Let $d \ge 2$ and $y \in \{\pm 1\}$. Fix, before seeing data, the following:*

- *a finite margin grid $\Gamma \subset (0, 8(d-1)]$;*
- *for $K \ge 0$, dyadic $B = 2^\ell$ and $\gamma \in \Gamma$, the grid steps $\eta_\lambda = \gamma/(8(d-1)\max(K,1))$ and $\eta_z = \gamma/(16(d-1)(B+1))$, a net $\mathcal{N}_{\eta_z}$ of S^{d-1}, and the class $\mathcal{G}_{K,B,\gamma} = \{\sum_{k\le K}\lambda_k' z_k z_k^\top : \lambda_k' \in \eta_\lambda\mathbb{Z},\ |\lambda_k'| \le \eta_\lambda\lceil B/\eta_\lambda\rceil,\ z_k \in \mathcal{N}_{\eta_z}\}$.*

The class size satisfies $C_{K,B,\gamma} := \ln|\mathcal{G}_{K,B,\gamma}| \le K\ln(2\lceil B/\eta_\lambda\rceil + 1) + dK\ln(1 + 2/\eta_z)$. *Put* $\epsilon_\gamma = \gamma/(8(d-1))$. *With probability* $\ge 1-\delta$, *simultaneously for all* $t \ge 0$, *all* $\gamma \in \Gamma$, *every integer* $N \ge N_{\epsilon_\gamma}(t)$ *and every dyadic* $B \ge P(t)$,

$$\operatorname{err}(\operatorname{sgn} f_t) \le \mathrm{kl}_+^{-1}\Big(\widehat{P}_S[yf_t \le \gamma],\ \frac{C_{N,B,\gamma} + \ln\frac{2|\Gamma|(N+1)(N+2)(|\ell|+1)(|\ell|+2)}{\delta}}{n}\Big).$$

The event does not depend on the dynamics, so the bound holds for every pair (U, V), *in particular for every GD or SGD iterate.*

Admissible choices.

- N: the measured $N_{\epsilon_\gamma}(t)$; the weight-free value $\lfloor \kappa(\tau/\Theta_{\epsilon_\gamma})^2 \rfloor$ of Theorem B.19(2), or of (5) for GD, which requires $\Theta_{\epsilon_\gamma} > 0$; or the trivial bound $\min(d, h) + \min(d, h')$.
- B: $P(0) + t\|y\|_n^2$ for the flow; the GD/SGD bound of Theorem B.19(5); or the measured $P(t)$, each rounded up to a dyadic value.

Sketch. Split $X = X_b + X_s$, where X_b keeps the eigencomponents of $UU^\top$ and $VV^\top$ of size $\ge \epsilon_\gamma$. Then $\operatorname{rank} X_b \le N$ and $\|X_b\|_* \le B$. Since $\|xx^\top - I\|_* = 2(d-1)$, the part X_s moves f by at most $\gamma/4$. Rounding X_b into $\mathcal{G}_{N,B,\gamma}$ costs less than $\gamma/4$ more. Transfer the margins and apply kl-Occam. The capacity is about $(d+1)N\ln(32d(B+1)/\gamma)$, i.e. 127–129 nats per unit at $d = 16$, $B = 1$, $\gamma = \frac{1}{2}$. The unrestricted degree-2 sign class costs about 964 nats there (Cover's count), so the rank bound wins only for $N \le 7$.

Numerical. [Numerical] Setup: $d = 16$, $h = h' = 16$, $\alpha = 10^{-6}$, GD with $\eta = 0.01$ and one SGD run, and exact test error over all 2^{16} inputs.

- *The inequalities hold.* The log-det clock holds to round-off (10^{-8}), and the rank bound is within a factor 1.7–2.7 of the truth.
- x_1x_2 *at* $n = 3000$. The bound is 0.084 with the measured rank 2, and 0.213 with the weight-free rank 4 and the proved GD norm bound of Theorem B.19(5). The unrestricted class gives 0.276, and the true error is 0. So 0.213 is a weight-free certificate for GD. The step-size condition holds along the run ($\max_j 2\eta\|M_j\|_{op} = 0.011$–$0.021$ over the replayed GD runs).
- *Where it fails.* On MAJ of three pairs the residual does not vanish, so τ keeps growing on the plateau. The weight-free rank bound grows from 23 to 47 while the true rank stays 6, and the D7 bound becomes vacuous (0.969). The same weakness affects every bound stated in effective time τ/Θ when the residual plateaus above zero.
- *SGD.* The proved SGD forms of Theorem B.19(5) give a rank bound of 181 at $t = 30$, above the trivial 32. With N capped at 32, the D7 bound is 0.89–0.90 from $t = 17$ on.

C Experimental details

Network training (`study/nn_train.py`).

- **Architecture.** $f(x) = a^\top \mathrm{relu}(W_2 \mathrm{relu}(W_1 x + b_1) + b_2)$, widths $N \times N$.
- **Initialization.** $W_1 \sim \mathcal{N}(0, 1/d)$, $W_2 \sim \mathcal{N}(0, 1/N)$, $a \sim \mathcal{N}(0, 1/N^2)$, zero biases.
- **Adam** [17] with $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and learning rate 10^{-3}. The per-layer multipliers, relative to base width 64, are 1 for W_1, b_1 and $64/N$ for W_2, b_2, a.

- **SGD:** learning rate 0.2, with multipliers $N/64$ for W_1, b_1, 1 for W_2, b_2 and $64/N$ for a.
- **Loss and data.** Square loss $\frac{1}{2}(f-y)^2$; minibatch 64, sampled with replacement from a fixed training set.
- **Runs.** Training sets are nested across n for each seed, and all (n, seed) networks are trained in parallel. The checkpoints are at steps 0, 10, 20, 50, 100, 200, 500, 1000, 2000, 3000, 5000, 7000, 10000, 15000 and 20000. The test set has 4096 fresh points.
- **Recorded quantities:** the network's Fourier coefficients on all characters of the relevant variables up to degree 4; the participation ratios of both layers' kernels; weight norms and displacements; and the test predictions.

Baselines.

- **NNGP kernel:** the infinite-width kernel of the same architecture at initialization, evaluated in closed form on the cube, used in kernel ridge regression with ridge $10^{-4}\,\mathrm{tr}K/n$, for $n \le 4096$.
- **Occam over juntas:** the smallest $k \le 4$ such that some k-subset of coordinates is consistent with the training data, predicting by its truth table, with unseen cells given the majority label. If no subset is consistent, the 4-subset with the fewest training mistakes is used.
- **sswm:** strong-selection/weak-mutation evolution of a single shared circuit of fan-in-2 gates (BUF, signed AND, XOR) with a ridge readout. One random local mutation is proposed per tick: adding one gate (weight 1), a chain of two or three fresh gates (weights ϵ, ϵ^2), or rewiring. Improvements are kept.

The race model (`study/models/race.py`). The settings are $\theta = 1.5$, ridge $\lambda = 10^{-3}$, integer weights in $\{-3, \dots, 3\}$, ReLU gates, proto-unit orders up to $K = 4$ with $P = 8$ units per order, 8 power-iteration steps per firing, and at most 48 gates. Full details and the 30 ablations are in `study/ABLATIONS.md`.

The minimal model and predictor (`study/models/minimal.py`, `study/minimal_predict.py`).

- **Gates.** Depth-1 gates have fan-in at most 4; depth-2 gates have fan-in at most 3, over active literals and earlier gates.
- **Dense units.** A linear Hebbian unit, and one order-2 spectral unit per stage.
- **Acceptance.** The acceptance rule is (1), with $c = 2.25$ and $a = 18$.

The hybrid model (`study/models/hybrid.py`). The race model plus an NNGP background in a joint ridge readout, with background ridge $c_{\mathrm{bg}} n/t$, $c_{\mathrm{bg}} = 0.003$, and growth stopped when the training MSE falls below 0.03. Above $n = 2048$ the background is a rank-512 random-feature approximation; this is used in the scoreboard but not in the diagnosis.

Compute. Everything ran on a 4-core CPU container. A model evaluation over the full (21 targets)$\times$ (11 n) $\times$ (3 seeds) grid takes a few to ten minutes single-threaded.

D How this paper was produced

This paper was written by an AI system, Claude (Anthropic), over several sessions on 23–24 October 2026. Kaarel Hänni prompted and steered it. No human has reviewed the proofs or the code.

Verification process.

- **Derivations.** Each of the four derivations behind Section 6, files `theory/A`–`D`, was read by an independent adversarial AI referee instructed to find errors. Each was revised, and then refereed a second time.
  - *Round 1.* Derivation A failed (one mislabelled claim, five major issues). B failed on two overclaims. C passed with three required revisions. D passed with minor issues.
  - *Round 2.* All four passed with only minor issues (7, 8, 10 and 7). These were then applied, and each set of fixes was checked by a further agent.
  - *Unrefereed addition.* The minibatch form of one item of D's Theorem D6(5) was added during that last step and has not itself been refereed.
  - *Records.* The referee reports are in `theory/checks/`.
- **Numbers.** The numbers in the summary write-up were checked against the generated data by independent fact-checking agents, and each flagged discrepancy was re-checked by a second agent before correction. Of 58 flags, 49 were confirmed and corrected.
- **Model write-ups.** The ablation, minimal-model and hybrid write-ups each had a separate verification pass that reproduced their numbers from the saved evaluations and corrected several mechanism claims.

Pre-registration. Two tests were run with predictions recorded before the network results were inspected: the $d = 120$ test of Section 4.3, and the composition test of Section 5. The file timestamps in the repository, and for the composition test the git history, show the order.

What remains unchecked.

- Literature claims rest on abstracts and memory of the papers, not on rereading them.
- The $d = 120$ section of the synthesis note has not had an independent verification pass beyond our recomputation of the networks' n^* and the timestamp check.

Reproducibility. All code is in `study/`: the network harness `nn_train.py` (JAX, CPU), the models in `models/`, and analysis scripts that write the generated reports (`*.md`). The training runs total a few CPU-hours. Raw network outputs were kept in a session scratch directory and are not in the repository; the scripts regenerate them.